Well-being in the Workforce: The Role of Career Crafting, Decent Work, PERMA+4, and Work Volition in Economically Marginalized (EM) Workers and Non-EM Workers

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Abstract

Well-being in the Workforce: The Role of Career Crafting, Decent Work, PERMA+4, and Work Volition in Economically Marginalized (EM) Workers and Non-EM Workers

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Well-being is the most widely studied topic in the field of organizational positive psychology, yet economic context is often overlooked despite approximately one-third of the U.S. working population being economically marginalized workers. Moreover, it is unknown whether different economic contexts produce gaps in necessary resources for career, employment, and consequently, well-being. To address this, the present study investigated career crafting, decent work, PERMA+4 and work volition and their relationship to well-being in two distinct economic groups: economically marginalized (EM) workers and non-EM workers.

A total of 629 full-time and part-time working adults aged 18 to 64, collected from an online survey, were included in the data analysis. First, to operationalize economically marginalized workers, theoretical aspects of economic resources were investigated and labeled as economic capital. Then the structure of economic capital and its relationship to well-being was examined. Confirmatory factor analysis, measurement invariance tests, and hierarchical regression analyses provided initial support for utilizing economic capital as a more comprehensive way to identify EM workers, above and beyond income. In the subsequent t-tests, it was found that EM workers (n= 176) reported significantly lower levels of well-being, decent work, PERMA+4, career crafting, and work volition than non-EM workers (n= 453). When the mediating role of decent work and PERMA+4 in the relationship between career crafting and well-being were examined, it was found that while decent work partially linked the relationship

among EM workers and non-EM workers, the mediating role of PERMA+4 on the relationship between career crafting and well-being was full and significant for both EM workers and non-EM workers. Moreover, among the nine building blocks of PERMA+4, only positive emotions, positive meaning, and growth mindset were significant variables that explained the relationship between career crafting and well-being in EM workers whereas all nine building blocks of PERMA+4 were significant mediators in the relationship between career crafting and PERMA+4 in non-EM workers. The moderating effect of age was not significant in the relationship between career crafting and decent work. Finally, a multi-group path analysis was performed to explore a more complex dynamic among the study variables. The improved model demonstrated structural invariance tests, and it was found that career crafting and work volition were significant predictors for decent work and PERMA+4, and decent work was also a significant predictor for PERMA+4, which in turn, predicted well-being. Unlike originally speculated, the mediating role of decent work disappeared, and PERMA+4 fully mediated the relationship between decent work and well-being in both EM and non-EM workers.

Study findings indicate that 1) economic capital can be better understood by overarching economic resources across past to future economic security than by income alone, and 2) EM workers possess significantly fewer career and life resources considered critical for a thriving life, implying inferior work and life conditions compared to those of non-EM workers, yet 3) the foundational roles of career crafting and work volition on both PERMA+4 and decent work is crucial across EM and non-EM workers, and finally 4) PERMA+4 is an encompassing and robust framework that enhances the predictive power for well-being consistent with prior research. Overall, this study contributes to the theory development of career crafting, PERMA+4, and the psychology of working theory by further revealing more complex relationships between

them. Additionally, this study contributes to practice by revealing the importance of tailored interventions and support systems for economically marginalized workers in promoting holistic well-being and positive functioning in the workplace.

Dedication

I dedicate this dissertation to my husband, Sanggyu, and my daughter, Jiseo, for their continued love, support, and patience throughout my years of study.

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Chapter 1: Introduction

Few would argue the importance of financial stability in experiencing happiness. While well-being stands as the most extensively studied construct in positive psychology (Ackerman et al., 2018; Kim et al., 2018), it's notable that many studies in organizational psychology have tended to neglect economic contexts concerning well-being, positing that money has a minimal impact on happiness. Even when considered, the emphasis often centers on upper-income individuals, with a focus on how to spend money well (e.g., Kahneman & Deaton, 2010). This may be true from the perspective of the non-linear relationship between income and well-being: the link between income and happiness is correlated up to a certain threshold, but beyond that point, the relationship plateaus (Inglehart et al., 2008; Kahneman & Deaton, 2010). A plateau point reported from a study is an income level of $80,000 (Kahneman & Deaton, 2010). Considering that 75% of working individuals earn less than $80,000 in the US (the U.S. Census data for 2022, as summarized on Wikipedia), it is not an inclusive argument. Moreover, depending on the source, approximately one-quarter or one-third of the U.S. working population are considered lower-income workers (Brown et al., 2020; Loprest et al., 2009). Therefore, ignoring such a large number of working adults whose income significantly correlates to their happiness brings about criticisms and critiques. One example is shown in van Zyl et al.'s (2023) systematic literature review stating, "positive organizational psychology can be driven by capitalist motives, aiming to commodify positivity and promote individualistic and consumeristic values within organizational settings" (p. 6). Ignoring marginalization is not solely a problem in the field of positive organizational psychology. In vocational psychology, Blustein (2006) argues that over the decades vocational psychology research has predominantly centered on privileged demographics who encounter fewer obstacles in their career path and are more inclined to find

meaning in their work compared to those without such privilege. As such, addressing research on the neglected economically marginalized population is more than imperative.

After the psychology of working theory (PWT) was proposed, research on marginalization has been rapidly increasing in the field of vocational psychology (e.g., Allan et al.; Duffy & Autin, 2013; Duffy et al., 2012;). While better-paying jobs are often depicted as a result of individual effort and ability, it is evident that individuals from low-income backgrounds often remain in low-paying jobs, perpetuating their low socioeconomic status and resulting in diminished well-being (Blustein, 2001; Duffy et al., 2016; Schoon & Polek, 2011), preceding a conclusion that an economically constrained context should be seen as a marginalizing condition (McWhirter & McWha-Hermann, 2021). In positive psychology, Donaldson and Donaldson (2020) first initiated the integration of economic resources into the essential building blocks of well-being. They focused on one's perception of feeling economically secure in making a living and found its positive relationship with well-being. Nonetheless, the vast majority of studies still utilize income as a variable when studying economically marginalized workers. The present study furthers this endeavor by consolidating the scattered pieces in economic constraints (Duffy et al., 2012), income, and economic security (Donaldson & Donaldson, 2021) into one, which is labeled as e*conomic capital* in this study, to define marginalized workers more comprehensively by examining their past, current, and future economics resources.

Building upon the understanding that economic disparity may play a critical role in well-being, solutions must extend beyond financial conditions. While the present study investigated the potential impact of economic stability on well-being variables, it also delved into the ways in which EM workers can attain improved employment. Decent work refers to work that provides a) adequate compensation, b) physical and psychological safety, c) access to healthcare, d)

adequate rest time, and e) value congruence between work and family/community values (Duffy et al., 2017; the International Labor Organization, 1999). In the psychology of working theory (PWT), the concept of decent work emerges as a critical framework. Originating from the United Nations Declaration of Human Rights (1948), decent work proposes five essential aspects of work conditions that are critical for human dignity. This includes not only adequate income but also emphasizes the importance of safety, health, work-life harmony, and work-life value congruence (Blustein et al., 2004; Duffy et al., 2016). A host of research on PWT makes it evident that decent work is a strong predictor of job satisfaction, subjective well-being, and mental health (see Buyukgoze-Kavas & Autin, 2019; Duffy et al., 2019; Isik et al., 2019; Kozan et al., 2019). As such, recognizing the vital role of decent work in enhancing well-being underscores a critical pathway toward improving the lives of EM workers.

Well-being cannot be understood without understanding the dynamics around work and career, particularly decent work (ILO, 2012). Decent work is characterized as work involving fair pay, safety, healthcare access, sufficient rest, and alignment with personal values (Duffy et al., 2017), and is considered a prerequisite for individuals' well-being (ILO, 1999). Research demonstrates that decent work is positively related to physical and mental health among diverse working adults, posing a significant role in their well-being (Blustein et al., 2001; Duffy et al., 2017; ILO, 2012). While many well-being studies focus on mental health, research has consistently shown a close relationship between mental health and career among working adults. A meta-analysis of 104 studies demonstrates that unemployment correlates with increased depression and poorer health outcomes, indicating that unemployed individuals experience lower psychological and physical well-being compared to their employed counterparts (McKee-Ryan et al., 2005). As such, it is crucial to recognize the intricate connection between well-being and

decent work, particularly when investigating how they relate to economically marginalized workers.

Decent work is crucial for enhancing well-being, yet achieving it poses significant challenges, especially for those facing economic disparities and systemic marginalization. Research indicates that contextual disadvantages—such as lifelong economic constraints and experiences of marginalization—severely impede the attainment of decent work, adversely affecting the well-being of working adults (Allan et al., 2020; Kozan et al., 2019). This situation prompts a critical inquiry: How can workers under economically scarce conditions navigate to experience greater well-being? Furthermore, while decent work encompasses essential elements for well-being, it often lacks specificity in how individual actions can directly contribute to achieving such well-being. To address this gap, the present study further investigated career crafting. By examining career crafting, the present study aimed to understand a more action-oriented and adaptable strategy that individuals can employ to enhance their well-being, despite their contextual challenges.

As the current study incorporates career crafting to add specificity in action to fill the gap left by decent work, it is crucial to understand the dynamics of career crafting and its potential impact on well-being. Career crafting emerges as a series of behaviors that individuals proactively engage in to shape their work and career paths in ways that align with their personal values, interests, and abilities (Ge et al., 2023; Lee et al., 2021; Tim et al., 2022). This approach is particularly relevant in the context of economic disparities and systemic marginalization, where traditional pathways to decent work and well-being may be hampered. Career crafting facilitates the establishment of relational, cognitive, and task resources. Through the lens of resource conservation theory (Hobfoll, 1988, 1989, 2011), career crafting may cultivate a

broader spectrum of personal and professional resources. These resources are instrumental in mitigating the adverse effects of economic scarcity and leveraging personal agencies toward achieving improved well-being (Hobfoll & Schumm, 2009). In essence, career crafting is an important tool to help with decent work, highlighting the individual's active role in navigating and overcoming the barriers to well-being. Economic context operates at the systematic societal level, whereas decent work functions at the organizational level, and career crafting occurs at the individual level. The present study investigates all these different levels as resources that may intertwine and address the well-being of EM workers.

By examining career crafting as a means to generate resources that help navigate better employment while dealing with economic challenges, this study utilizes the PERMA+4 model as a theoretical framework to identify various positive states generated during the process of engaging in these crafting activities. Proposed by Donaldson and colleagues (2022), the PERMA+4 model offers comprehensive paths—from positive emotions and engagement to financial security and an eco-friendly work environment—which are essential for fostering a sense of well-being. By enhancing resources through career crafting, individuals may enrich their well-being across these varied domains amidst economic difficulties. Altogether, the present study investigated the relationships among career crafting, decent work, PERMA+4, and well-being, particularly seeking to understand how career crafting contributes to decent work and positive psychological states and, thus, to greater well-being, specifically in the economic marginalization context. Addressing the well-being of economically marginalized workers provides insight into the issue of societal economic inequality on a larger scale.

Chapter 2: Literature Review and Hypotheses Development

The purpose of Chapter Two is to review the literature and gain an understanding of critical resources related to well-being. The review begins by identifying economic marginalization and subsequently, the literature on decent work and career crafting is examined, with a focus on the context of economic marginalization. Finally, a comprehensive discussion of career crafting is provided, analyzing its relationship to the elements of well-being using the PERMA+4 framework.

Marginalization

Marginalization is defined as a state and a process whereby an individual or group is hindered from fully engaging in the social, economic, and political aspects of life (Alakhunova et al., 2015). Marginalization often occurs due to individuals' social and/or cultural underrepresentation, such as racial minorities, sexual minorities, people with low socioeconomic status, the disabled, and the like. Individuals in marginalized conditions tend to experience more social exclusion, feel less important in society, and have less access to resources. (Blustein, 2001; Duffy et al., 2020; Eldering & Knorth, 1998). A host of studies revealed that marginalization negatively related to decent work, job satisfaction, life satisfaction, and meaningful work, to name a few (Allan et al., 2020; Douglass et al., 2017; Duffy et al., 2018; Wang et al., 2019).

Lee (2021) conducted a systematic literature review on marginalized individuals' occupational trends using the database of the Web of Science to understand how marginalization is related to career. Lee (2021) used "the psychology of working" (topic) as the search term, which resulted in 51 articles, including 37 empirical studies. Lee (2021) then included the terms "career (title)" and "minority OR under-represented* OR vulnerable OR disadvantaged OR

marginalized (title)” to capture articles about marginalized individuals' careers. The search result initially identified 194 articles. After removing non-psychology-related fields such as engineering and economics, 28 empirical articles remained. By synthesizing the identified articles, Lee identified three themes critical to marginalized individuals' career trajectories.

Lacking Relational Resources

Relational resources like family are keys to opening career opportunities for decent work (Blustein, 2011; Kenny & Medvide, 2013). Marginalized individuals lack professional role models because their parents lack a strong network. Family background is critical when searching for a job because it influences young adults' network quality (Schoon & Heckhausen, 2019). Some studies indicate that parents of a lower socioeconomic status are less interested in their child's career, which leads to the child's lower career aspirations (Masdonati & Fournier, 2015; Schoon & Polek, 2011). Other studies reported that youths with immigrant backgrounds had fewer opportunities for decent employment and positive career development than their native-born peers and less access to employment when leaving school (Ng-Knight & Schoon, 2017; Yakushko et al., 2008). As such, marginalized individuals lack career mentoring or coaching to help guide them to find decent work.

Marginalized individuals' lack of relational resources is also pertinent to a lower level of educational attainment compared with other populations. A study found that youths with immigrant backgrounds have higher school dropout levels than their native-born peers (Ali & Menke, 2014). Educational attainment predicts the quality of career paths (Blustein et al., 2002; Ling & O'Brien, 2013; OECD, 2017; Rojewski & Kim, 2003; Vuoloet al., 2014). Given that education provides a quality network that the family members of marginalized individuals cannot

offer, lower educational attainment attenuates opportunities to establish a more robust network for career exploration toward decent work.

Lacking Psychological Resources

The second theme identified in Lee's (2021) study is a lack of psychological resources. Research indicates that the experience of marginalization negatively influences one's confidence to explore career options and shrinks one's "can do" mindset. For example, research constantly shows that marginalized individuals have a low level of work volition. Work volition refers to the perception of one's capacity to choose their career freely (Duffy et al., 2012). For instance, even in the face of external obstacles, when someone believes that they are finding their own path in the working world instead of feeling compelled to take any job they can find due to their financial situation, they exhibit a higher level of work volition. Work volition predicts greater access to decent work, job satisfaction, a stronger perceived fit with one's work environment, and greater perceived meaning at work (Duffy et al., 2018, 2020; Heckhausen, 2019). Moreover, Kim and Lee (2020) found that a lower level of work volition is related to a lower level of a positive state of self, such as lower self-esteem, self-efficacy, and locus of control. These studies suggest that lacking psychological resources, such as work volition and positive self-evaluation, hampers marginalized individuals' career exploration of decent work.

In addition, Vuolo et al. (2014) found that youths of low socioeconomic status exhibited less competent and smooth transitions from school to work. Furthermore, Masdonati and Fournier (2015) reported that youths of low socioeconomic status scored lower on career aspirations than youths of higher socioeconomic status. According to Schoon and Polek's (2011) longitudinal study of a large sample in the British population, career aspirations at their teenage years predicted career success (i.e., defined as net income and occupational status in the study) in

their mid-30s. Altogether, these studies on marginalization offer evidence that a lack of positive psychological resources will likely undermine marginalized individuals in finding decent work.

Lacking Career Clarity

Lee (2021) identified that a lack of vocational clarity can explain the hindrance of marginalized individuals' access to decent work. Ng-Knight and Schoon (2017) found that young adults with lower socioeconomic status showed a less crystallized self-concept than peers of higher socioeconomic status. Being aware of vocational interests, abilities, and skills contributes to setting and achieving career goals (Bandura & Locke, 2003). Thus, vocational clarity is important for achieving career success but is a missing element for marginalized individuals. Research further indicates that women struggle more to link their vocational identity with decent work. For instance, Schoon and Polek (2011) reported that young women transitioning from school to work experienced less access to higher-paying occupational sectors. One explanation for this gender imbalance in decent work is that women have little information and few role models to guide them about previously male-dominant occupations (e.g., engineering; Imdorf et al., 2015). As such, structural perceptions limit women's career choices for decent work. Together, the previous findings suggest that marginalized individuals may struggle more to picture their career identity than non-marginalized individuals, especially in associating their career with decent work.

In summary, Lee (2021) concluded that the marginalization literature revealed that marginalized individuals' career development is more challenging than that of non-marginalized individuals. The literature identifies three barriers that hinder marginalized individuals' ability to gain greater access to decent work in their careers: a) a lack of relational resources that help guide them on how to navigate decent work, b) a lack of psychological resources that provide

confidence to explore decent work, and c) a lack of vocational clarity that provides a clear goal to help gain decent work. These barriers for marginalized individuals can be changed (Duffy et al., 2019). The present study introduces career crafting and work volition as two malleable variables and looks at their relationship with decent work in the economic marginalization context.

Economic Marginalization

This study defines economic marginalization as a status of lacking economic resources that affect one's psychological and physical state. Positing that economic resources can be better understood by assessing past experiences, present income, and future outlooks rather than relying solely on current income, this section reviews three levels of economic resources: income, past economic constraints, and future economic security. The psychology of working theory (Blustein, 2001; Duffy et al., 2012) and economic security from PERMA+4 (Donaldson & Donaldson, 2021) mainly aided in shaping this approach.

Income

Lower income is used as a common indicator when identifying economically unfavorable conditions (Ross & Mateman, 2019). Considering multiple government reports, income less than approximately $30,000 can be considered as lower income. One approach to classifying lower income is to use the federal poverty level (FPL). The U.S. Census Bureau classifies workers whose income level does not exceed 200% of FPL as lower income (Semega & Kollar, 2022). Considering FPL is $14,580 in 2023, individuals' income that is lower than $29,160 per year can be considered low income. According to the Kaiser Family Foundation (2022), approximately one-quarter of US workers fall into the low-income category. Another approach to classifying low income is to use below two-thirds of the median hourly wage for full-time workers.

According to the U.S. Bureau of Labor Statistics (May 2020), the median hourly wage for full-time workers in California, for example, was $24.64 per hour and thus approximately $16.43 per hour and below ($31,545 when converted to yearly full-time wage) can be considered low income. According to the Government Accountability Office (2017), approximately 37 percent of families with a worker fall into this lower wage category. Both approaches indicate that a significant number of individuals are considered economically marginalized workers (EM).

Lacking adequate income increases the likelihood of experiencing stress due to a lack of buffer against unexpected life events such as illness (Diener et al., 2013; Tay & Diener, 2011). A meta-analytic study of 86 randomized controlled trials with working-age individuals between 16-64 years old across countries found that the effects on well-being are greater when income changes cross the subsistence threshold, although the exact threshold varies across studies (see Thomson et al., 2022). As such, an adequate level of income is a crucial factor in defining financial resources.

Past Economic Constraints History

While income is commonly employed as a measure of financial resources, a significant body of literature suggests that assessing one's past economic constraints and future perception of economic security can be a way to assess one's economic marginalization more comprehensively than income alone.

One factor to address past economic constraints is to look at one's lifetime history of economic constraints. Duffy et al. (2019) conceptualized it and defined it as an individual's subjective experience of financial instability throughout their life (Duffy et al., 2019). As opposed to viewing income as a consequence of individuals' efforts and abilities, growing literature recognizes that income inequality relates to their previously marginalized economic

context. According to the psychology of working theory (PWT), social and economic conditions negatively affect one's confidence to freely choose careers, which in turn limits access to work that provides better compensation and work environments (Blustein, 2001; Duffy et al., 2016). Multiple studies indicate that individuals with lower-income jobs have previously experienced economic constraints. For example, one study reports that children from low-income families exhibit lower career aspirations than others (Masdonati & Fournier, 2015). Career aspirations are critical for career success. Schoon and Polek's (2011) longitudinal study reveals that career aspirations during participants' teenage years are shown to be a critical predictor for more net income and higher occupational status in their mid-30s. Multiple studies suggest that this is because children from low socio-economic backgrounds have poor career networks (Schoon & Heckhausen, 2019), lack career clarity (Ng-Knight & Schoon, 2017), have a much higher ratio of school dropouts than other children (Ali & Menke, 2014), and less confidence in transitioning from school to work (Vuolo et al., 2014). All of these factors contribute to fewer opportunities for decent employment and positive career development (Ng-Knight & Schoon, 2017; Yakushko et al., 2008). When considered collectively, these points suggest that EM may have faced greater lifetime economic constraints in their past and present.

Future Economic Security

Whereas economic constraints pertain to the experience of economic scarcity in the past, economic security refers to one's perceived financial ability that extends to the future (Donaldson et al., 2022). Therefore, it is worthwhile to consider economic security when assessing an individual's economic capital. Economic security involves having adequate assets other than their income. When they lose several months of pay due to serious illness, workers who have savings to support themselves tend to feel economically secure (Donaldson & Donaldson, 2020).

Suppose a worker's current annual income is less than $30,000, and they have no savings, assets, or available support (i.e., someone willing to give or lend them money) to sustain themselves for several months without a job. In this case, they are considered low-income workers and economically insecure. On the other hand, consider another worker whose current annual income is the same as the worker above (i.e., less than $30,000), but they have savings, assets, or support (such as a wealthy parent) to rely on for several months without a job. Although these workers also are low-income, they are economically secure. The ability to handle future events can influence one's attitudes toward their career, life, and overall well-being. Economic security also involves the capacity to plan, manage, and control spending over their income, rather than the actual amount of income. A higher level of economic security enables individuals to make bigger life plans and decisions and increases well-being (Donaldson et al., 2022; Gill et al., 2001; Margolis et al., 2021; Sieg & Wang, 2017). As such, economic security plays a crucial role in comprehending one's economic conditions.

To summarize, it is proposed that understanding economic resources using past, current, and future perspectives is a more comprehensive and accurate way to gauge one's economic condition and its impact on their physical and psychological state. To name this conceptual variable, I labeled it as *economic capital* and hypothesized it to be a comprehensive representation of one's economic resources, encompassing past, present, and future aspects.

H1: Past economic constraints, present income, and future economic security compose economic capital.

Well-being

In this study, well-being refers to an individual's accumulated overall life satisfaction (Diener, 1985), which is the most common understanding in well-being literature to date. Many

decades ago, it was concluded that higher happiness was related to demographic factors such as being a “young, healthy, well-educated, well-paid, extroverted, optimistic, worry-free, religious, married person with high self-esteem, job morale, modest aspirations, of either sex and of a wide range of intelligence" (Wilson, 1967, p. 294). This view has been greatly challenged by subjective well-being theory with a great emphasis on psychological factors (Diener, 1984). Moreover, it was emphasized that subjective well-being necessarily includes positive elements that transcend economic prosperity (Diener et al., 1999).

Vast investigations into subjective well-being have provided a deeper understanding of important psychological elements related to happiness, and the role of economic factors has been downplayed. For example, a meta-analysis of 49 independent studies (Shin & Lyubomirsky, 2009) reported several moderators, including age and participants' depression status, but economic factors were excluded. If it existed, the focus was materialism rather than the actual economic status (e.g., Dittmar et al., 2014).

However, economic factors should not be neglected because they are critical for one’s well-being, particularly for working adults. A meta-analysis of 104 studies reported that unemployment correlates with increased depression and poorer health outcomes, revealing that unemployed individuals experience lower psychological and physical well-being compared to their employed counterparts (McKee-Ryan et al., 2005). While considering the cruel impact of economic factors on one’s physical and psychological well-being, it is unjust if economic factors are excluded in well-being studies that might impact one’s psychological state and, thus, well-being.

Research consistently shows that economic status is a critical factor in determining one's well-being. To be specific, studies indicate that the relationship between income and well-being

is linear up to a certain point, beyond which further income gains have a diminishing effect (Inglehart et al., 2008; Kahneman & Deaton, 2010). One explanation is that, as Diener et al. (2018) note, lacking financial resources means there is safety net to rely on during unexpected life events, such as illness or daily hassles. Lacking financial resources also limits the ability to obtain goods and services that satisfy their needs and thus negatively impacts one's well-being (Tay & Diener, 2011). As such, workers lacking economic capital are more vulnerable to stressful life events, which can have a negative impact on their overall well-being (Tay et al., 2018). Therefore, it is evident that economic capital is an essential resource when it comes to achieving higher levels of well-being.

Career Crafting

Career crafting refers to a set of proactive behaviors invested in advancing, modifying, and altering career boundaries to make one's career congruent with their emerging self (Lee et al., 2021). Research demonstrates that career crafting was positively related to meaningful work, work engagement, and subjective career success, performance, and person-job fit (Chifor & Oprea, 2023; Lee et al., 2021). The underlying premise of career crafting is that successful lifelong careers involve both proactivity and congruence in their careers: Individuals seek congruence between a career and their values, needs, and interests, driving proactive behavior (Tims et al., 2012; Wrzesniewski & Dutton, 2001) and career congruence (Hall, 1976; Mainiero & Sullivan, 2006).

Proactivity

Career crafting is characterized by behaving proactively and seeking congruence between work and personal interests, values, and strengths. Proactive behavior involves stepping out to "make things happen rather than watching things happen" (Parker et al., 2010, p. 828). Parker et

al. (2010) define proactivity with three attributes: self-starting, change-oriented, and future-focused. Through this lens, career crafting behaviors such as taking actions to initiate relational and task career resources (self-starting), being open to opportunities to develop their career (change-oriented) and making their work more meaningful (future-focused) are examples of proactivity. Research suggests that proactive people are more innovative, engage in greater career initiatives, and develop more political knowledge (Seibert et al., 2006). Thus, career crafters who are taking proactive steps enjoy more effective acquisition and preservation of resources.

Congruence Seeking

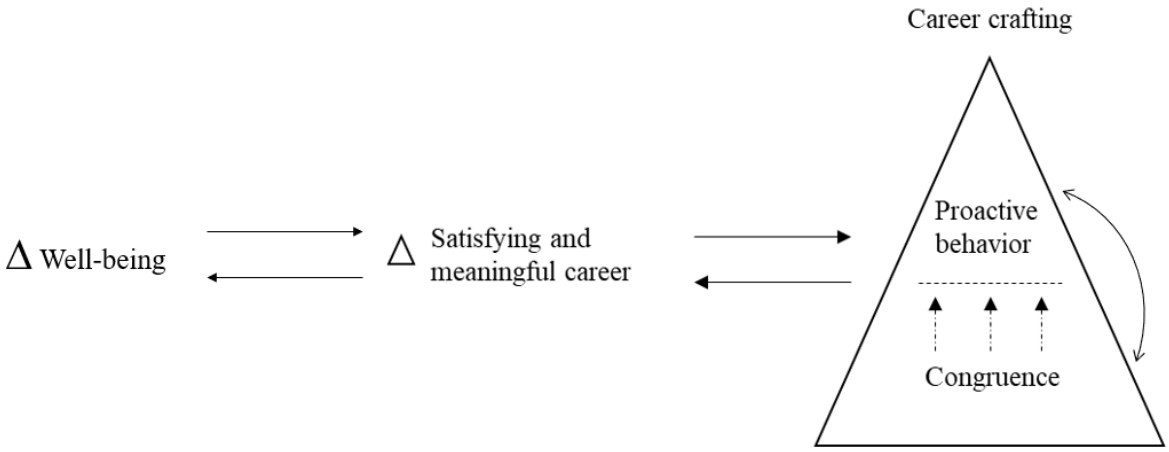
Another characteristic of career crafting reflects the need for congruence between one's job, values, and authentic self for career satisfaction (e.g., Hall, 1976; Mainiero & Gibson, 2018). As Parker et al. (2016) suggest, career proactivity should be personally meaningful to be sustainable. Studies indicate the more workers perceive their career path aligns with their internal guide, the more satisfied they are with their careers. For example, when workers find their work similar to their self-concept (i.e., personality, values, and interests), they assess their work more positively, which increases job satisfaction, work engagement, and performance (e.g., EMards, 1991; EMards & Shipp, 2007; Kristof-Brown, 1996; Verquer et al., 2003). This suggests that career crafting builds internal resources by satisfying the need for congruence between self and career, which relates to improved well-being.

According to the conservation of resources (COR) theory, individuals seek to obtain and preserve various resources, including objects, psychological characteristics, conditions, and energies, because lacking resources produces a variety of negative emotions, such as anxiety, anger, or depression (Hobfall 1988, 1989, 2002). These resources are interconnected and

influence one's perception of resource gain or loss as a total. COR theory further suggests that acquired resource reserves can offset losses (Hobfoll, 1988, 1989, 2011; Hobfoll & Schumm, 2009). For example, one study found that women who experienced substantive gains were significantly less affected by losses than those who had experienced no resource gain (Wells et al., 1997, 1999). Therefore, although inadequate financial resources may have a negative impact on well-being, engaging in career-crafting behaviors can potentially facilitate the acquisition of other types of resources that can have a positive effect on overall well-being. Figure 1 presents the conceptual background of career crafting.

Figure 1

The Conceptual Model of Career Crafting[1].



Note. Individuals seek congruence between a career and their values, needs, and interests, driving proactive behavior. This leads individuals to increase their career satisfaction and meaning in work, reinforcing congruence seeking.

[1] This model was the improved version of the one initially developed by Lee et al. (2021)'s unpublished manuscript.

When acquiring resources, career crafters tend to focus on four key areas: changing relational boundaries, utilizing relational resources, reflecting positive meaning, and expanding task boundaries (Lee et al., 2021).

Relational Crafting (Changing Relational Boundaries and Utilizing Relational Resources)

Relational crafting in one's career is defined as "the extent to which individuals proactively search for and connect with a group of people with whom they can share their authentic interests and values." (Lee et al., 2021, p. 719). Two types of relational crafting were identified in their study: changing relational boundaries and utilizing relational resources.

Changing Relational Boundaries. Changing relational boundaries involves developing new social resources by connecting with individuals who share career interests. Research demonstrates that social resources positively correlate with current salary, the number of promotions throughout a career, rapid promotion, and career satisfaction (Seibert et al., 2001; Shipilov et al., 2017). Career crafters establish stronger connections with people who share their career interests and possess the skills they aim to develop. These informal networks provide access to information, resources, and career sponsorship, which, in turn, enhance career success (Seibert et al., 2001).

Utilizing Relational Resources. Utilizing Relational Resources involves proactive resource/information-seeking behavior utilizing social networks. This form of crafting is characterized by several behaviors, such as seeking professional coaching from individuals whose careers are admired, attending events that offer insights into various career paths, requesting introductions to influential people within one's professional network, and soliciting feedback on career development. These actions represent a proactive effort to harness the power

of social networks and mentorships, reflecting a deliberate investment in one's career growth and development (Lee et al., 2021). Studies indicate that obtained information influences future career behaviors when it is proactively utilized (Stumpf et al., 1983), suggesting that career crafters who seek resources and information shape their careers to be more congruent to their evolving self and thus find their careers more meaningful.

Cognition Crafting (Reflecting Positive Career Meaning)

Reflecting positive meaning is a form of cognitive crafting, which is defined as the degree to which “individuals proactively reflect on career meaning and frame their career as an essential part of their meaningful life" (Lee et al., 2021, p. 719). Career crafters view their career not merely as a means of living but as a vital component of life, interpreting one's job as a part of fulfilling their life story (Wrzesniewski & Dutton, 2001). Jobs with low income tend to be repetitive and uninteresting (Reggio, 2017). However, when individuals perceive their career as an essential part of their meaningful life, it increases meaning and thus well-being. Additionally, because career crafters seek congruence between their interests and their work, they remain in or find jobs that allow them to experience meaning (Lee et al., 2021). Studies indicate that the more workers perceive their career path as aligning with their internal guide, the more satisfied they are with their careers. For instance, research indicates that jobs such as hospital cleaning (Wrzesniewski & Dutton, 2001) and portering (Fuller & Unwin, 2017) are considered low-income, repetitive, and uninteresting. Career crafters find meaning and thus are more satisfied with their work.

Cognitive reframing might not be achievable when a job is initially chosen that is misaligned with one's interests, values, and abilities. Career crafters seek careers congruent to their internal attributes and resources. According to the person-job fit theory, when workers

perceive their work as a reflection of their self-concept (i.e., personality, values, and interests), they are likely to view their work more positively. This positive evaluation enhances job satisfaction, work engagement, and performance (e.g., EMards, 1991; EMards & Shipp, 2007; Kristof-Brown, 1996; Verquer et al., 2003). As such, enhancing meaning is not separate from pursuing careers more congruent to oneself, and when both are present concurrently, it is possible for the job to be more meaningful and satisfying, even in low-income careers.

Task Crafting (Expanding Task Boundaries)

Expanding task boundaries is a form of task crafting, which is defined as "the extent to which individuals develop skills and abilities that lead to the realization of their best self within their career" (Lee et al., 2021, p. 719). Typical behavior is choosing to take on additional tasks at work that contribute to their career whether they are paid tasks or not. They do not impose these tasks on themselves; rather, they undertake them because they find enjoyment and excitement in doing so.

Expanding task boundaries can also be understood through the literature on career competencies, which identifies essential knowledge, skills, and abilities as crucial elements for career advancement, and they are within individuals' capacity to influence and develop (Akkermans et al., 2013). Research indicates that increases in career competencies are positively associated with increases in job resources, work engagement, employee health, job satisfaction, and career success (Akkermans & Tims, 2017; Akkermans et al., 2013; Plomp et al., 2016).

To summarize, career crafting offers a holistic approach to examining career actions. According to multiple reports, working adults change their jobs an average of 12 times or every 4.5 years until they retire (the Bureau of Labor Statistics, 2018; Gratton et al.; Walsh & Volini, 2017). Whether these changes are voluntary or involuntary, such frequent job transitions

necessitate an ongoing career strategy throughout one's work lifespan. By being aware of congruence as an internal compass and embodying it through proactive career behaviors involving relational, cognitive, and task crafting, career crafters are likely to acquire internal and external resources. These resources enable them to navigate the dynamic context of the job market, guiding them toward fulfilling short-term and long-term satisfying careers.

Decent Work

The psychology of working theory (PWT) was developed to understand how people gain access to decent work and how having decent work impacts their work-related and overall well-being (Duffy et al., 2016). The PWT posits that access to decent work significantly enhances both work-related and overall well-being, fulfilling fundamental human needs for survival, social contribution, and self-determination (Autin et al., 2019; Duffy et al., 2016; 2020).

Decent work is a concept with the purpose of promoting the realization of employee rights and interests in terms of overall human dignity including freedom, equality, and security. Decent work is the core concept of the PWT believed to be a prerequisite for individuals' well-being (International Labor Organization [ILO], 1999). Although decent work was initially proposed by ILO as a global movement, it is not unrelated to the U.S., considering socioeconomic disparities and the unequal opportunities to obtain quality employment are great concerns within its labor market. Decent work is “the sort of work that ought to define the lives of all who work and who want to work” (Blustein et al., 2016, p. 1). It is the guideline for the quality of work that people should be able to access in contemporary society (ILO 2014, 2015). The concept of decent work originates from the United Nations Declaration of Human Rights (1948), which contains the following statement, emphasizing work as an integral aspect of human rights:

> Everyone has the right to work, to free choice of employment, to just and favorable conditions of work and to protection against unemployment. (2) Everyone, without any discrimination, has the right to equal pay for equal work. (3) Everyone who works has the right to just and favorable remuneration ensuring for himself and his family an existence worthy of human dignity, and supplemented, if necessary, by other means of social protection. (4) Everyone has the right to form and to join trade unions for the protection of his interests.

Decent work was more thoroughly defined to enable statistical investigation by Duffy and colleagues (2017). They characterized decent work as employment that provides a) adequate compensation, b) physical and psychological safety, c) access to healthcare, d) adequate rest time, and e) value congruence between work and family/community values (Duffy et al., 2017).

Decent work has increasingly been recognized as a critical factor in contributing to well-being. Research suggests that decent work, rather than just income, is crucial for improving social and economic status as well as overall well-being. Numerous studies conducted across the globe, since the decent work scale was developed in 2017, have shown strong relationships to job satisfaction, meaningful work, and well-being (see Allan et al., 2020; Douglass et al., 2017; Duffy et al., 2019; Wang et al., 2019). Moreover, these studies have revealed that decent work is positively associated with other desired organizational outcomes, such as person-organization fit, and negatively associated with withdrawal intentions (Buyukgoze-Kavas & Autin, 2019; Isik et al., 2019). Overall, these findings highlight the importance of work that satisfies fundamental human needs in contributing to a more comprehensive and critical experience of well-being beyond obtaining economic resources.

The relationship between decent work and well-being may also be applicable in the economic marginalization context. Research has shown that decent work is positively related to mental and physical health in lower-income workers and ethnic minorities. For instance, Duffy et al.'s study (2019) reported that individuals with an annual household income under $50,000 who are employed in decent work experience better mental health, while meeting their survival needs contributes to better physical health.

Career Crafting and Decent Work

Career crafting is relatively independent of economic marginalization, whereas decent work is closely related to it. Research indicates that economic constraints are negatively related to accessing decent work. Kozan et al. (2019) report that social class influences decent work both directly and indirectly through factors such as work volition (i.e., the degree of freedom in career choice) and career adaptability, which, in turn, is related to job satisfaction and overall life satisfaction. Similarly, Masdonati et al. (2019) reveal that perceived social class and qualitative job security are significant predictors of decent work, with work volition partially mediating these relationships. Furthermore, Kim and Lee’s (2020) study finds that low work volition, which demotivates individuals to seek decent work, is related to low self-esteem, self-efficacy, and locus of control. These relationships indicate that economic marginalizations can undermine individuals' confidence in choosing their career, potentially leading to a decreased ability to secure decent work. Overall, these studies suggest that economic status negatively impacts their attainment of decent work.

Career crafting is less likely to be influenced by economic obstacles. For example, Hirschi and Freund's (2014) longitudinal study found that career barriers did not inhibit participants' career engagement. According to the COR theory, resources must be invested to

gain or preserve them (Hobfoll, 1989), and career crafting can be seen as an act of creating a resource gain cycle. Low-income jobs often lead to limited access to career-related resources, such as training opportunities and social networks that provide job-related information and resources (Blustein, 2001). Career crafting can increase these critical career resources for EM independently from economic constraints, making it an empowering tool to offset their lack of resources, which ultimately may increase well-being.

Furthermore, while decent work informs essential resources individuals gain from work to maintain human dignity, such as salary, health insurance, and rest time, it does not explain the informative aspects of obtaining these resources such as how to obtain essential work resources. In contrast, career crafting provides a pathway to generate critical career resources such as how to obtain decent work and how the process can produce other types of resources (relational, task, cognitive resources) that can reduce the negative impact of a lack of finances. According to practice theory (Bourdieu, 1990), a career is sustained when individuals continuously carry out career-related actions, such as networking, learning, and reframing (Greene & Rau, 2018). Careers unfold through the ongoing performance of career actions rather than individuals controlling and planning their careers (Forrier, 2023). As such, the ongoing enactment of activities to sustain social phenomena applies to career, and career crafting plays a role as a continuing process of career enactment that can build resources from one activity to another, broadening and building a career repertoire (Fredrickson, 2018). Therefore, this review speculates that career crafting generates resources that uniquely contribute to well-being.

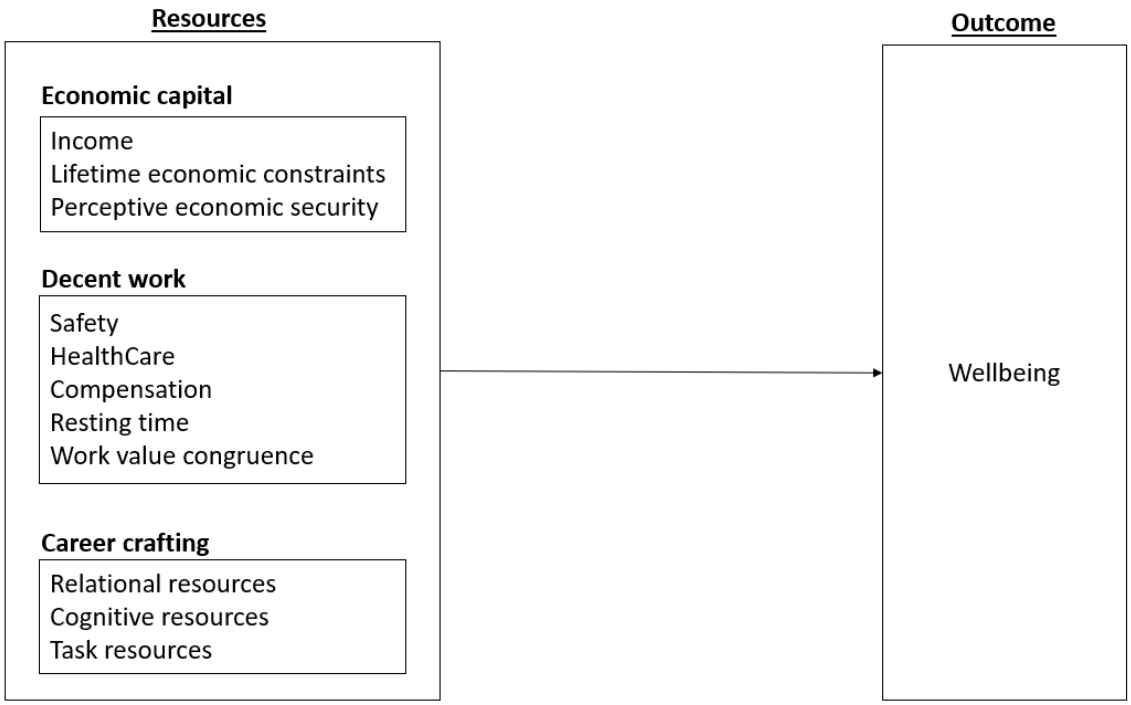
To summarize, this review identifies economic resources, career crafting, and decent work as three distinct predictors contributing to the well-being of working adults. Economic capital serves as a resource for meeting financial needs, whereas decent work provides a

resourceful employment context that supports human dignity. Career crafting, on the other hand, involves proactive actions that build a process of accumulating necessary resources. These three variables are speculated to contribute to well-being. Figure 2 represents economic capital, career crafting, and decent work as resources to predict well-being.

H2. Economic capital, decent work, and career crafting uniquely contribute to well-being.

Figure 2

Economic Capital, Decent Work, and Career Crafting as Unique Predictors for Well-being



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Work Volition

The psychology of working theory (PWT) proposed two antecedents for decent work: work volition and career adaptability. Work volition refers to one's perception of their capacity to freely choose their career, capturing a feeling of choice in one’s career decision (Duffy et al.,

2012; 2020). Research has found that work volition was a strong predictor for positive work and personal outcomes in various contexts. For example, a study with sexual minority employed adults (Douglass et al., 2017) and two studies with racial and ethnic minority employed adults (Duffy et al., 2018; 2019) found significant relationships between work volition and decent work. A longitudinal study with three time points over six months (Duffy et al., 2020) also found significant relationships between work volition and decent work. This evidence suggests work volition is a critical predictor of decent work.

Research indicates that work volition is affected by economic context. A study found that youths of low socioeconomic status exhibited less confidence in transitioning from school to work (Vuolo et al., 2014) and scored lower on career aspirations than youths of higher socioeconomic status (Masdonati & Fournier, 2015). Kozan et al. (2019) report that social class influences decent work both directly and indirectly through work volition, which in turn, is related to job satisfaction and well-being. Kim and Lee (2020) explain that low levels of self-esteem, self-efficacy, and locus of control are related to low work volition and low motivation to seek decent work. All this evidence suggests that although work volition is an essential psychological state for achieving greater decent work, the starting point may be unequal for economically marginalized individuals: contextual disadvantages can impair their ability to exercise their sense of control over career choices.

It is worth noting that another suggested antecedent in the PWT is career adaptability. Savickas (2002) defined it as “a psychological construct that denotes individuals’ readiness and resources for coping with current and anticipated tasks of vocational development” (p.156). However, subsequent research has shown an insignificant relationship between career adaptability and decent work (e.g., Douglass et al., 2019; Duffy et al., 2020). Thus, while career

adaptability has been excluded from later studies, work volition consistently stands out as a significant predictor of decent work across multiple studies.

Career crafting is likely to be independent of economic context, making it an empowering tool to offset the lack of financial resources, which may contribute to one's access to decent work. While work volition is an attitude that is linked to social status, career crafting involves taking actions to acquire tangible career-related resources, such that career crafters acquire resources by exploring tasks that may enhance their skills (task crafting), building networks by asking for connections to those who interest them (relational crafting), and viewing their behaviors as a quest for meaning in their lives (cognition crafting; Lee et al., 2021). COR theory suggests that acquired resource reserves can offset losses (Hobfoll, 1988, 1989, 2011; Hobfoll & Schumm, 2009). For example, one study found that women who experienced substantive gains were significantly less affected by losses than those who had experienced no resource gain (Wells et al., 1997, 1999). As such, resources generated by engaging in career crafting may help one acquire decent work, even after accounting for work volition and, thus, well-being.

H3. Controlling for work volition, career crafting positively contributes to the variance in decent work, which in turn contributes to variance in well-being in the economic marginalization context.

Worker Age as a Moderator

Multiple studies imply that worker age may moderate the relationship between career crafting and decent work. For example, a study (Li et al., 2021) found that employees' age moderates the indirect effects of job crafting on job satisfaction through person-job fit such that the connection between task and relational crafting and person-job fit is stronger among younger workers, whereas the relationship between cognitive crafting and person-job fit is stronger

among older workers. The study discusses that the difference may be because younger workers are more motivated to access decent work, while older workers prioritize work meaningfulness. Another study reports that the relationship between job crafting and work engagement weakens significantly in older workers. Similar to job crafting, career crafting may have a more substantial impact on younger adults seeking to enter decent work, as they form critical resources for decent work through career crafting while older workers may have already accumulated necessary resources through their years of work experience. Thus, it is speculated that when both young and older workers engage in the same level of career crafting, younger workers may perceive a greater level of decent work compared to older workers. Therefore, it is possible that worker age moderates the relationship between career crafting and decent work such that the relationship is stronger among younger workers than older workers in EM. Figure 2 presents the hypothesized study model, path analysis among career crafting, work volition, worker age, decent work, PERMA+4, and well-being.

H4. Worker age moderates the relationship between career crafting and decent work such that the relationship is stronger among younger workers than older workers in EM.

PERMA+4

One goal of researchers studying subjective well-being is to define the key features of the well-being construct (Kozma et al., 1991). Considerable effort has been invested in integrating components of both hedonic and eudaimonic well-being into the study of well-being, emphasizing the presence of positive psychological resources (e.g., Diener, 1984; Ryan & Deci, 2001; Ryff, 1989). Seligman's (2011) PERMA model presented a sophisticated approach that outlined paths to achieving well-being. PERMA represents five positive psychological states including positive emotions, engagement, relationships, meaning, and accomplishments.

Seligman clarifies that PERMA is a precursor to well-being, arguing that PERMA's value lies in its framework for enhancing well-being through interventions aimed at bolstering each of the PERMA components. A recent systematic review of a large volume of empirical studies on PERMA (Cabrera & Donaldson, 2023) confirms that PERMA is a positive and strong predictor for well-being.

PERMA+4 is a stronger predictor for well-being than PERMA. Seligman (2018) suggests that PERMA is not exhaustive and thus additional building blocks of well-being are needed to be established. Building upon Seligman's calling for an expansion of PERMA, Donaldson and Donaldson (2020) introduced PERMA+4, by adding four additional components—economic security, positive health, growth mindset, and positive work environment—into PERMA, establishing nine building blocks of well-being. While PERMA primarily addresses the psychological states and experiences of well-being, PERMA+4 provides a more comprehensive framework, particularly valuable in work settings, demonstrating that PERMA+4 is a positive predictor for reduced turnover intentions, higher levels of adaptivity, proactivity, and proficiency for both self-report data and co-workers' reported data (Donaldson & Donaldson, 2020; Donaldson et al., 2021). Moreover, PERMA+4 is an encompassing and robust framework that enhances the predictive power for well-being (Cabrera & Donaldson, 2023). The details of each PERMA+4 component are outlined in Table 1 and are discussed in the following section.

Table 1

Definitions of the Building Blocks of PERMA+4 (Donaldson & Donaldson, 2020, pp.4-5)

Building Block	Definition
Positive Emotions	Experiencing happiness, joy, love, gratitude, etc., here and now.

Positive Engagement	Being highly absorbed, immersed, or experiencing flow while engaged in activities of one's life.
Positive Relationships	Having the ability to establish and maintain positive, mutually beneficial relationships with others characterized by experiences of love and appreciation.
Positive Meaning	The experience of being connected to something larger than the self or serving a bigger purpose.
Accomplishment	Experiencing a sense of mastery over a particular domain of interest or achieving important or challenging life/work goals.
Positive Health	Operationalized as a combination of high levels of biological, functional, and psychological health assets.
Growth Mindset	A mindset characterized by an optimistic, future-oriented view of life, where challenges or setbacks are seen as growth opportunities.
Positive Work Environment	The quality of the physical work environment (which includes spatiotemporal elements, such as access to natural light, fresh air, physical safety, and a positive psychological climate) aligned with the preferences of the individual.
Economic Security	Perceptions of financial security and stability required to satisfy individual need.

PERMA+4 As a Mediator

Several components of PERMA+4 can explain the relationship between career crafting and well-being. Table 2 summarizes the potential relationships between career crafting and each element of PERMA+4.

Positive Emotions. Donaldson and Donaldson (2020) describe positive emotions as the present moment experiences of feelings such as happiness, joy, love, and gratitude. While engaging in career crafting, workers may find joy and happiness because they experience increased task abilities, have favorable work boundaries, build meaningful relationships, and

have personal meaningfulness for their career journey, which may be lacking otherwise. Studies indicate that crafting career boundaries produce favorable changes in both tangible and intangible work role boundaries, such as the experience of positive emotions (Bruning & Campion, 2018; Lichtenthaler & Fischbach, 2019; Tims et al., 2013). Increased positive emotions may also motivate EM to continue, creating a bidirectional gain cycle (Hobfoll, 1989). Furthermore, research suggests that acquiring career resources helps enhance self-efficacy. Self-efficacy has been demonstrated as a source of positive feelings (Jerome et al., 2002) and positive affect (Rego et al., 2012). From all of these, it is speculated that there is a positive relationship between career crafting and positive emotions.

Positive Engagement. Positive Engagement refers to the state of being highly absorbed, immersed, or experiencing flow while engaged in activities of one's life (Donaldson & Donaldson, 2020). In terms of the relationship between career crafting and engagement, the second component of PERMA+4, research shows that challenges play a critical role in fostering engagement, especially when tasks are characterized as boring (Nakamura & Csikszentmihalyi, 2009). Tasks for low-income workers are often repetitive and tiring, in contrast to higher-income jobs that tend to be more creative and complex. Career crafting introduces challenges that facilitate personal growth and enhance the overall learning experience. Multiple studies have reported that increased crafting at work, such as taking on new projects (e.g., Harju et al., 2016; Lee et al., 2021; Tims et al., 2013), relates to higher levels of work engagement. Therefore, individuals who engage in career crafting are more likely to experience higher levels of engagement at work.

Positive Relationships. Positive relationships are about the capacity to form and sustain beneficial connections with others that are marked by feelings of love and appreciation, and

mutual positivity (Donaldson & Donaldson, 2020). Career crafting may also be closely related to positive relationships. Research indicates that relational crafting is at the core of career crafting (Lee et al., 2021; Tims & Akkermans, 2020). Career crafters actively seek and connect with individuals in their career context who share their authentic interests and values. These evolving boundaries in the quantity and quality of interactions with people, both inside and outside work, may contribute to the enhancement of positive relationships. This is especially crucial in the context of the EM, where individuals may lack a strong network or mentor (e.g., Schoon & Heckhausen, 2019). The act of seeking and connecting through career crafting has the potential to cultivate positive relationships.

Positive Meaning. Meaning is a fundamental component of overall well-being, as highlighted by research indicating that a greater sense of meaning is associated with higher levels of well-being (Peterson et al., 2005; Seligman, 2011; Steger et al., 2011; Triplett et al., 2012). Meaning in work and life is about fostering eudaimonic well-being, which emphasizes personal growth, utilizing one's strengths, and finding purpose (Dik et al., 2015). Positive meaning is about the experience of being connected to something larger than the self or serving a bigger purpose (Donaldson & Donaldson, 2020).

Understanding meaning can be broken down into three concepts: coherence, purpose, and significance (Martela & Steger, 2016). Coherence is the sense that life is logical and consistent; purpose refers to having clear goals and a sense of direction; and significance is the belief that one's life is valuable and worth living. To be specific to a work context, understanding meaning in work is also broken down into three aspects: reflecting positive meaning in career, meaning making through work, and greater good motivation (Steger et al., 2012). When workers' overarching beliefs and goals align with a subjective sense of meaning, this alignment not only

shapes how they interpret their experiences and work towards their aspirations but also influences their understanding of day-to-day experiences, thereby fostering a sense of meaningfulness and motivating them to avoid unhappiness by reinterpreting situations in meaningful ways (Park & Ai, 2006).

Career crafting may also contribute to positive meaning, the fourth component of PERMA+4. Cognitive crafting is one of the fundamental domains within career crafting. Career crafters engage in reflective practices to discern the significance of their careers, interpreting them as a means to find greater meaning in their lives (Lee et al., 2021). Low-income work may pose more barriers to finding meaning in work. For instance, a study found that temporary workers perceive their crafting efforts as less impactful and receive fewer benefits than permanent workers, which demotivated them to proactively change their work boundaries (Plomp et al., 2019). Consequently, discovering work-related meaning may require intentional cognitive processes, particularly in low-income contexts. For example, in Wrzesniewski and Dutton's (2001) qualitative study, a cleaning lady in a hospital who exhibited higher job crafting found greater meaning in her work. All of this suggests that cognitive crafting can create a more meaningful work experience, contributing to a greater sense of meaning in work.

Positive Accomplishment. Positive accomplishment is about experiencing a sense of mastery over a particular domain of interest or achieving important or challenging life/work goals (Donaldson & Donaldson, 2020). Career crafting may also relate to a sense of accomplishment, the fifth component of PERMA+4. Career crafting is a series of behaviors that help one acquire relational, task, and cognitive resources, which is essential for career development. Multiple studies suggest that gaining these resources is positively related to experiencing more accumulated positive psychological outcomes from work (Cenciotti et al.,

2017; Lee et al., 2021; Tims & Akkermans, 2020). This implies that career crafting contributes to a greater sense of accomplishment at work.

Positive Health. Positive health is operationalized as a combination of high levels of biological, functional, and psychological health assets (Donaldson & Donaldson, 2020). It is a crucial component of both well-being and mental health (Cross et al., 2018; Donaldson et al., 2022). When individuals proactively engage in establishing career-related resources, this may increase one's confidence and control over their bodies. However, it is also possible that one's perception on their physiological functioning may influence one's actions modifying and broadening career boundaries. Although the positive relationship between career crafting and positive health is speculated, because little evidence exists in the field of organizational studies, the relationship between career crafting and positive health will be better understood after further exploration in this study.

Growth Mindset. Growth mindset is characterized by an optimistic, future-oriented view of life, where challenges or setbacks are seen as growth opportunities (Donaldson & Donaldson, 2020). It is likely that career crafting is positively related to mindset, which is the seventh component of PERMA+4. Individuals with a growth mindset typically choose more demanding tasks that can challenge their current abilities because they hold positive beliefs that their work will provide them with opportunities for growth and meaningful challenges to stretch their capabilities (Donaldson et al., 2020; Donaldson & Donaldson, 2021; Dweck, 2006, 2016). Career crafting and a growth mindset may establish a positive *gain cycle* (Hobfoll, 1989, 2011), similar to the relationship between career crafting and positive emotions. Obtaining resources through career crafting can increase positivity and optimism, which, in turn, can motivate individuals to proactively and continuously engage in career crafting.

Positive Work Environment. The quality of the physical work environment (which includes spatiotemporal elements, such as access to natural light, fresh air, physical safety, and a positive psychological climate) aligned with the preferences of the individual (Donaldson & Donaldson, 2020). It includes the extent of access to natural light, fresh air, physical safety, and a positive psychological climate (Donaldson et al., 2020; Sander et al., 2019). Low-income work conditions often entail precarious work conditions (Blustein et al., 2019). Although career crafting may help individuals identify and pursue opportunities for better work conditions, its contribution to the work environment may not be strong in low-income contexts.

Economic Security. Economic Security refers to perceptions of financial security and stability required to satisfy individual need (Donaldson & Donaldson, 2020). It involves an individual's ability to plan, manage, and control their spending in relation to their income (Donaldson et al., 2022). While career crafters may exhibit effective career planning and resource management, there is no clear evidence connecting career crafting to economic security. From the review of potential relationships between career crafting and the nine components of PERMA+4, a hypothesis for the positive relationship between career crafting and overall PERMA+4 is established. Table 2 illustrates the hypothetical relationships between career crafting and PERMA+4 among EM. As such, career crafting may have relations with several of these PERMA+4 components, which strengthens paths to well-being.

Table 2

The Hypothetical Relationships between Career Crafting and PERMA+4 among EM Workers

PERMA+4	Potential links to Career crafting

Positive Emotions	In the process of establishing career resources, career crafters obtain various resources that may produce positive emotions such as happiness and joy.
Positive Engagement	While a task is seen as an opportunity to grow their career rather than seeing it as merely things to do for a living, career crafters may tend to be more engaged in their work.
Positive Relationships	In the process of establishing career resources, career crafters may have more opportunities to develop relationships with those who share their career interests.
Meaning	Career crafters view their career to be aligned with their life goals and part of the quest of their life journey.
Accomplishment	Career crafters are open to changing the boundaries of their job, and putting their effort into making things happen. This attitude would lead to a higher level of accomplishment at work.
Positive Health	Career crafters are active in viewing their lives, which may relate to one's confidence and control over their bodies.
Grow Mindset	Career crafters are hopeful for future careers and may view current challenges as an opportunity to grow.
Work Environment	Career crafting may be independent of contextual conditions.
Economic Security	Career crafting may be independent of contextual conditions.

From the potential relationships between career crafting and several elements of PERMA+4, it is speculated that career crafting generates some of these positive psychological states, which, in turn, relate to greater well-being.

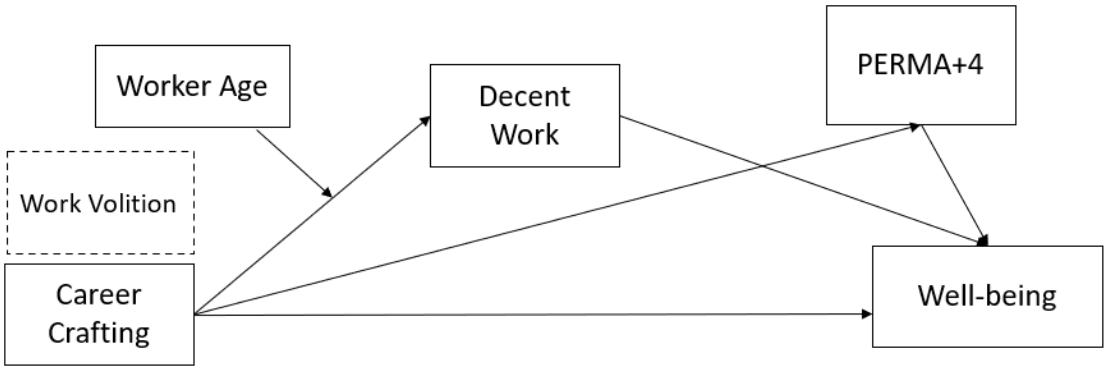
H5. Career crafting positively contributes to variance in overall PERMA+4, which in turn contributes to variance in well-being among economically marginalized workers.

Proposed Entire Paths

Building upon the hypothesized relationships and their respective directions, I have established a path model. This model aims to investigate whether career crafting and work volition serve as two critical predictors for the concepts of decent work and PERMA+4. In turn, these elements are posited to be significant predictors of well-being, incorporating all the possible dynamics among the study variables, including worker age. Figure 3 illustrates the entire set of path relationships among the variables under study.

Figure 3

Proposed Entire Paths Among Study Variables



Note. Career crafting and work volition are two predictors for decent work and PERMA+4, which in turn positively relate to increases in overall well-being.

Chapter 3: Method

The present study utilized a cross-sectional correlational design, which is appropriate because the primary aim was not to claim causality but to examine the degree and patterns of

relationships among the study variables, which are not detectable using other research designs (Crano et al., 2011; Spector, 2019). This study also utilized self-reported data. Although some caution is needed when interpreting the study results, self-reported measures are generally reliable (Diener et al., 2002; Donaldson et al., 2021). In addition, because the study variables did not include sensitive questions that could discomfort respondents, the influence of social desirability was relatively minimal (Krumpal, 2013).

Participant

Participants were recruited using Connect, a crowd-sourcing online research platform. While originally, 530 participants were planned to be recruited based on the power estimation, following MacCallum et al.'s (1996) suggestion, due to an insufficient number of EM group participants recruited from the initial data collection, an additional 150 participants were recruited, resulting in a total of 680 participants. After removing unqualified participants (five responses with a short time spent on the survey of less than three minutes, 13 responses due to unqualified employment status, three univariate outliers, and 30 multivariate outliers), 629 participants were entered for data analysis.

Of the entire sample (N = 629), 55% were male (n = 345), 45% were female (n = 280), and 1% were others (n = 4). The ages of the participants ranged from their 20s to their 60s, with 86% falling within the 20s to 40s age range. In terms of education, 20% (n = 127) indicated they are high school graduates or an equivalent, 15% (n = 96) indicated they possess a 2-year degree, 48% (n = 301) indicated they possess a 4-year degree, 16% (n = 103) indicated they possess a master's degree or higher. Regarding income, 27% (n = 169) indicated that their income is below USD 40,000, 33% (n = 205) reported between USD 40,001 and USD 70,000, and 23% (n = 144) reported above between USD 70,001 and USD 100,000, and 17% (n = 111) reported USD

100,001 and above. Regarding work tenure, 89% (*n* = 561) of the sample indicated that they have more than five years of work experience in a full-time capacity. Finally, regarding race/ethnicity, 70% indicated that they are White/Caucasian (*n* = 439), followed by African American/Black (12%, *n* = 77), Asian (6%, *n* = 40), Hispanic/Latino (4%, *n* = 28), and American Indian/Alaskan Native (1%, *n* = 4). 6% (*n* = 40) identified themselves as Multiracial. Table 3 presents the characteristics of the entire sample, EM workers, and non-EM workers.

Table 3

Sample Characteristics

		Total		EM Workers		Non-EM Workers	
		n	%	n	%	n	%
Gender	Male	345	55%	75	43%	270	60%
	Female	280	45%	99	56%	181	40%
	Other	4	1%	2	1%	2	0%
Age	Age 20~29	123	20%	44	25%	79	17%
	Age 30~39	277	44%	73	41%	204	45%
	Age 40 ~49	141	22%	40	23%	101	22%
	Age 50 ~59	63	10%	15	9%	48	11%
	Age 60+	24	4%	4	2%	20	4%
Education	High school and equivalent	127	20%	62	35%	65	14%
	2-year degree	96	15%	44	25%	52	11%
	4-year degree	301	48%	55	31%	246	54%
	Master's degree +	103	16%	15	9%	88	19%
Work Year	Less than 2 years	26	4%	6	3%	20	4%
	3-4 years	42	7%	17	10%	25	6%
	5-6 years	74	12%	24	14%	50	11%

	7-8 years	59	9%	19	11%	40	9%
	9-10 years	108	17%	30	17%	78	17%
	11 years+	320	51%	80	45%	240	53%
Income	Less than $30,000	68	11%	56	32%	12	3%
	$30,001 - $40,000	101	16%	59	34%	42	9%
	$40,001 - $60,000	138	22%	44	25%	94	21%
	$60,001 - $80,000	127	20%	14	8%	113	25%
	$80,001 - $100,000	84	13%	3	2%	81	18%
	$100,001 - $150,000	71	11%	0	0%	71	16%
	$150,001 +	40	6%	0	0%	40	9%
Employment Status	Full-time paid	584	91%	150	85%	434	96%
	Part-time paid	45	7%	26	15%	19	4%
Race/ Ethnicity	American Indian/Alaskan Native	4	1%	2	1%	2	0%
	Asian	40	6%	4	2%	36	8%
	Black or African American	77	12%	27	15%	50	11%
	Hispanic/Latino	28	4%	7	4%	21	5%
	Native Hawaiian or Pacific Islander	0	0%	0	0%	0	0%
	White or European Caucasian	439	70%	127	72%	312	69%
	Multiracial	40	6%	9	5%	31	7%

Note. EM =Economically Marginalized Workers. Sample size Total =629, EM=167, and non-EM= 458.

Procedure

An online survey, formulated via Qualtrics, was administered to collect data from participants on Connect, an online crowdsourcing platform. This survey comprised a consent form, measurement items for study variables, an attention check, and demographic questions. Upon selecting the survey title, “Work, Career, and well-being Survey,” participants were first shown the consent form, and only those who consented were allowed to proceed. An attention check was placed midway to ensure participant engagement. Demographic details such as age, gender, educational attainment, income, years of work experience, and ethnicity were collected at the end of the survey to minimize the impact of decreasing motivational levels (Crano et al., 2011). Upon completion, respondents were directed to a debriefing page explaining the study's objectives and providing a completion code for compensation.

Data were collected two times. The initial collection targeted 530 participants without imposing income criteria. While this met the sample size requirements for the first and second hypotheses of the study, only 15% had an income range of less than USD 50,000. Anticipating the potential lack of representation for EM, an additional 150 samples were collected from individuals earning less than USD 50,000, utilizing the targeting function on the Connect platform. The first 530 respondents who completed the survey received a USD 2.30 compensation. The additional respondents were paid USD 2.00, reflecting the finding that the average survey duration was shorter than initially estimated (see Buhrmester et al., 2011).

Measures

Economic Capital

Economic capital was assessed using questions from three scales: five questions from the economic constraints scale (ECS; Duffy et al., 2019), three questions from economic security, a domain within PERMA+4 (Donaldson & Donaldson, 2021), and a single question regarding

annual income. economic capital utilizes a seven-point Likert scale (1 = s*trongly disagree* to 7 = s*trongly agree*). A sample item from ECS includes, "For most of my life, I have not felt financially stable." A sample item from Economic Security includes, "I could lose several months of pay due to serious illness and still have my economic security." Yearly income was assessed with a single question: “Which range of amounts best describes your current estimated yearly income?” The response set for the annual income originally provided 15 choices, ranging from “Less than $10,000” to “$150,000 and above,” which were later recategorized into seven points. In prior research, the validity of the ECS was supported by data from a sample of employed racial/ethnic minorities, showing moderate correlations with social status, annual income, subjective social class, and decent work (Duffy et al., 2019). The internal consistency of economic capital in the present study sample was α = .93. See Appendix A for the entire set of items.

Well-being

The 5-item satisfaction with life scale (SWLS, Diener et al., 1985) was used to measure individuals' subjective well-being. The response set is a seven-point Likert scale (1 = s*trongly disagree* to 7 = s*trongly agree*). A sample item includes "In most ways, my life is close to my ideal." This scale was initially validated using data from a sample of undergraduate students (Diener et al., 1985) and later validated using data from peer reports, a memory measure, and clinical ratings (Pavot et al., 1991). In the present study sample, the internal consistency was α = .93. See Appendix A for the entire set of items.

Career Crafting

The 15-item career crafting assessment (CCA, Lee et al., 2021) was used to measure career crafting. The CCA comprises four factors: changing relational boundaries, utilizing

relational resources, reflecting positive career meaning, and expanding task boundaries. The response set is a seven-point Likert scale (1 = d*oes not describe me at all* to 7 = c*ompletely describes me*). Sample items from each dimension include: "I make connections with people who share my career interests." (changing relational boundaries), "I seek professional coaching from those whose careers I admire." (utilizing relational resources), "I think about the ways in which my career positively impacts my life." (reflecting positive career meaning), and "I choose to take on additional tasks at work." (expanding task boundaries). The validity of CCA was supported by data from a diverse sample of working adults (Chifor & Oprea, 2023; Kilic & Kitapci, 2024; Lee et al., 2021). In the present study sample, the internal consistency was α = .94. See Appendix A for the entire set of items.

Decent Work

The 15-item decent work scale (DWS, Duffy et al., 2017) was used to measure decent work. The DWS comprises five three-item subscales corresponding to the five factors of decent work, which include adequate health care, safe working conditions, adequate compensation, hours that allow for free time and rest, and organizational values that complement family and social values. The response set is a seven-point Likert scale (1 = *strongly disagree* to 7 = s*trongly agree*). Sample items include “My employer provides acceptable options for health care” (access to health care), “At work, I feel safe from emotional or verbal abuse of any kind” (safe working conditions), "I am rewarded adequately for my work" (adequate compensation), “I have free time during the work week” (hours that allow for free time and rest), and “The values of my organization match the values within my community” (organizational values complement family and social values). The DWS was found to be correlated with job satisfaction, work meaning, and

withdrawal intentions (Duffy et al., 2017). In the present study sample, the internal consistency was α = .89. See Appendix A for the entire set of items.

PERMA+4

I utilized 29 items from the nine domains of the positive functioning at work Scale (PF-W, Donaldson & Donaldson, 2020): positive emotions, engagement, positive relationships, positive meaning, positive accomplishment, positive health, economic security, growth mindset, and positive work environment. The PF-W is rated on a 7-point Likert-type scale (1 = *strongly disagree* to 7 = s*trongly agree*). PF-W was validated using multi-trait, multi-method data from a sample of working adults recruited via an online survey (Donaldson & Donaldson, 2020; Donaldson et al., 2021). The internal consistency in the present study sample was α = .93. See Appendix A for the entire set of items.

Worker Age

The worker ages were categorized into six scales (1 = *19 and under* to 6 = *60 and above*).

Work Volition

The 14-item work volition scale (WVS; Duffy et al., 2012) was used to measure work volition. The response set is a seven-point Likert scale (1 = *strongly disagree* to 7 = s*trongly agree*). A sample item includes "I've been able to choose the jobs I wanted." Prior research indicates that the WVS correlates with job satisfaction, work locus of control, core self-evaluations, and decent work (Douglass et al., 2017; Duffy et al., 2012; 2016). Internal consistency in the present study sample was α = .92. See Appendix A for the entire set of items.

Chapter 4: Analysis and Results

Data Cleaning and Preliminary Analysis

I used R (version 2023.12.0) for the data analysis. Data were inspected to identify unqualified data. Thirteen data points were removed due to unqualified employee status (those who identified themselves as self-employed, student, retired, or homemaker). Five responses were removed due to an unrealistic survey completion speed (under three minutes). Negatively worded items were reverse-coded. Mean scores for each variable were created. Income was recorded from the originally collected 11 levels into seven levels to ensure that all measures have a range from one to seven.

Then data were inspected to identify if there are missing data, univariate outliers, and multivariate outliers. There was only one missing data point, which was replaced by the mean value. Three data points were identified as univariate outliers, based on the over ± 3 SD rule, and thus removed. Thirty multivariate outliers were removed when applying the Mahalanobis distance, applying a $p < .001$ and the corresponding Chi-squared (χ^2) value with the degrees of freedom equal to the number of variables. As a result, 629 data points were included in the primary data analysis.

The normality of the data was inspected through skew (±3 ≥) and kurtosis (±10 ≥) using descriptive statistics (Kline, 2015). Data linearity was confirmed from the visual inspections of the GG plots and QQ plots. No patterns emerged from the residuals, satisfying the assumption of homoscedasticity. However, multivariate normality was violated ($p \geq .05$) with the Henze-Zirkler test. Thus, robust maximum likelihood estimation was used for relevant analyses.

Descriptive Statistics and Correlation Matrix

Descriptive statistics, Cronbach alphas, and correlations among study variables can be found in Table 3. Most of the variables were positively moderately correlated with one another. Career crafting and subcomponents of career crafting were positively strongly correlated, decent

work and subcomponents of decent work were also positively strongly correlated, and overall PERMA+4 and each of the PERMA+4 building blocks were positively significantly correlated. Internal consistency tested with Cronbach's Alpha and McDonald's Omega demonstrated adequate reliability for each measure except Environment ($\alpha = .65$). Although it was slightly below what is considered to be acceptable, the overall PERMA+4 score exhibited strong internal consistency ($\alpha = .93$). Table 4 presents the means, standard deviations, correlation coefficients, and internal consistencies for the study variables and their respective subscales.

Table 4

Means, Standard Deviations, Cronbach Alphas, and Correlations Among Study Variables

	M	*SD*	α	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1 Well-being	4.34	1.55	.92																						
2 Career crafting	4.54	1.32	.94	.34***																					
3 Rel. Expansion	4.73	1.53	.93	.30***	.87***																				
4 Rel. Utilization	3.97	1.68	.89	.28***	.86***	.77***																			
5 Career Meanin	4.89	1.47	.89	.30***	.83***	.61***	.56***																		
6 Task Expansio	4.55	1.64	.88	.27***	.80***	.53***	.52***	.63***																	
7 Decent Work	4.94	1.04	.89	.47***	.39***	.32***	.31***	.39***	.31***																
8 Safety	5.86	1.11	.81	.24***	.23***	.18***	.13***	.26***	.20***	.63***															
9 Healthcare	5.12	1.78	.97	.29***	.29***	.26***	.25***	.26***	.21***	.69***	.29***														
10 Compensation	4.16	1.63	.88	.40***	.29***	.25***	.25***	.26***	.22***	.75***	.35***	.38***													
11 Rest	4.70	1.54	.86	.27***	.12**	.07	.07	.16***	.08*	.67***	.32***	.27***	.40***												
12 Value	4.85	1.46	.95	.40***	.42***	.32***	.34***	.40***	.37***	.70***	.41***	.31***	.42***	.32***											
13 PERMA+4	4.95	.91	.93	.58***	.62***	.50***	.45***	.61***	.54***	.71***	.50***	.40***	.54***	.41***	.64***										
14 Positive Emoti	4.56	1.58	.93	.46***	.61***	.48***	.44***	.59***	.55***	.58***	.36***	.24***	.44***	.34***	.64***	.82***									
15 Engagement	5.09	1.29	.86	.20***	.35***	.23***	.18***	.37***	.41***	.25***	.24***	.09***	.15***	.11***	.31***	.53***	.45***								
16 Relationships	5.29	1.22	.91	.32***	.42***	.32***	.30***	.42***	.37***	.58***	.56***	.28***	.40***	.35***	.49***	.72***	.58***	.30***							
17 Meaning	5.14	1.55	.95	.37***	.57***	.47***	.40***	.62***	.45***	.45***	.31***	.28***	.26***	.20***	.54***	.71***	.68***	.38***	.48***						
18 Accomplishme	5.64	.97	.76	.36***	.47***	.41***	.28***	.46***	.43***	.42***	.35***	.20***	.30***	.22***	.41***	.68***	.52***	.43***	.49***	.51***					
19 Health	5.29	1.18	.84	.40***	.19***	.15***	.14***	.18***	.17***	.43***	.36***	.23***	.33***	.32***	.28***	.56***	.33***	.12***	.34***	.21***	.33***				
20 Environment	4.67	1.35	.65	.28***	.34***	.29***	.29***	.29***	.26***	.42***	.29***	.26***	.29***	.28***	.33***	.56***	.36***	.15**	.34***	.24***	.24***	.35***			
21 ES	3.86	1.83	.89	.54***	.24***	.21***	.22***	.19***	.18***	.48***	.20***	.38***	.49***	.26***	.27***	.57***	.29***	.09*	.28***	.18***	.23***	.40***	.28***		
22 Mindset	5.01	1.38	.88	.47***	.53***	.45***	.40***	.50***	.46***	.62***	.41***	.34***	.51***	.35***	.54***	.82***	.70***	.33***	.57***	.54***	.49***	.35***	.41***	.45***	
23 Work volition	4.80	1.24	.92	.50***	.31***	.26***	.17***	.33***	.29***	.60***	.43***	.34***	.47***	.44***	.39***	.66***	.48***	.24***	.48***	.41***	.43***	.45***	.34***	.51***	.58***

Note. N = 629. Rel. = Relational.

* $p < .05$, ** $p < .01$., *** $p < .001$.

Hypothesis 1

The statistical adequacy of the structure of economic capital was tested to determine if past, present, and future economic conditions together represent economic marginalization. Three theoretically identified factors were entered as sub-factors of economic capital: past economic constraint (Factor 1), present income (Factor 2), and future economic security (Factor 3). Factor 1 consists of five items from the lifetime economic constraints Scale (Duffy et al., 2019). Factor 2 includes one item from the income question, and Factor 3 consists of three items from economic security (Donaldson & Donaldson, 2020). All items were measured using 7-point scales. Income was re-coded from the originally collected 16 levels into seven levels. See Appendix B for entire items and their factor loadings.

Table 5 presents the means, standard deviations (SDs), Cronbach's alphas, and correlations among the three factors for economic capital. The data demonstrated adequate internal consistency for economic capital and its sub-factors: internal consistencies for all items were $\alpha = .93$ for economic capital, .89 for Factor 1 (past economic constraints), 1.00 for Factor 2 (income), and .96 for Factor 3 (future economic security), respectively. The correlation coefficients among the sub-factors showed positive, significant, and moderate relationships with one another ($p < .001$).

Table 5

Means, SDs, Cronbach's alphas, and Correlations among the Factors for Economic Capital

	Measure	*M*	*SD*	α	1	2
1 Past Economic Constraints	Lifetime Economic Constraint Scale (ECS, Duffy et al., 2019)	3.77	1.82	.96		
2 Present Income	Income	3.69	1.7	1.00	.40***	
3 Future Economic Security	Economic Security (Donaldson & Donaldson, 2020)	3.86	1.84	.89	.56***	.47***

Note. N = 629. All were measured using 7-point scales. Items of past economic constraints were reverse-coded.

Confirmatory Factor Analysis

Model fit was examined using confirmatory factor analysis with the robust maximum likelihood estimation procedure to determine whether the theoretically identified economic capital can demonstrate an appropriate statistical structure. I compared models fit among a one-factor model, a three-factor model with correlated factors, and a higher-order factor with a three-factor model. The results are presented in Table 6. I first established a one-factor model, which resulted in poor fit. Then, I ran the three-factor model with correlated factors. Finally, I performed a higher-order factor with a three-factor model. The fit indices indicated a good model fit: CFI =.99, Chi-square to degrees of freedom ratio (χ^2/df) = 2.87, RMSEA = .06, and SRMR = .03, and 90% CI = [.04, .07] (Hu & Bentler, 1999; Kline, 2015; MacCallum & Austin, 2000). Finally, I performed a higher-order factor with a three-factor model. The three factors loaded solidly onto the higher-order factor, ranging from .58 to .83. AIC was inspected to determine the better fit; however, the values of AIC were identical (AIC = 18305) attributed to the same degrees of freedom. This indicates that both models are statistically adequate for explaining the variance structure in economic capital, thereby supporting Hypothesis 1. Figure 4 represents a higher-order factor with a three-factor model.

Table 6

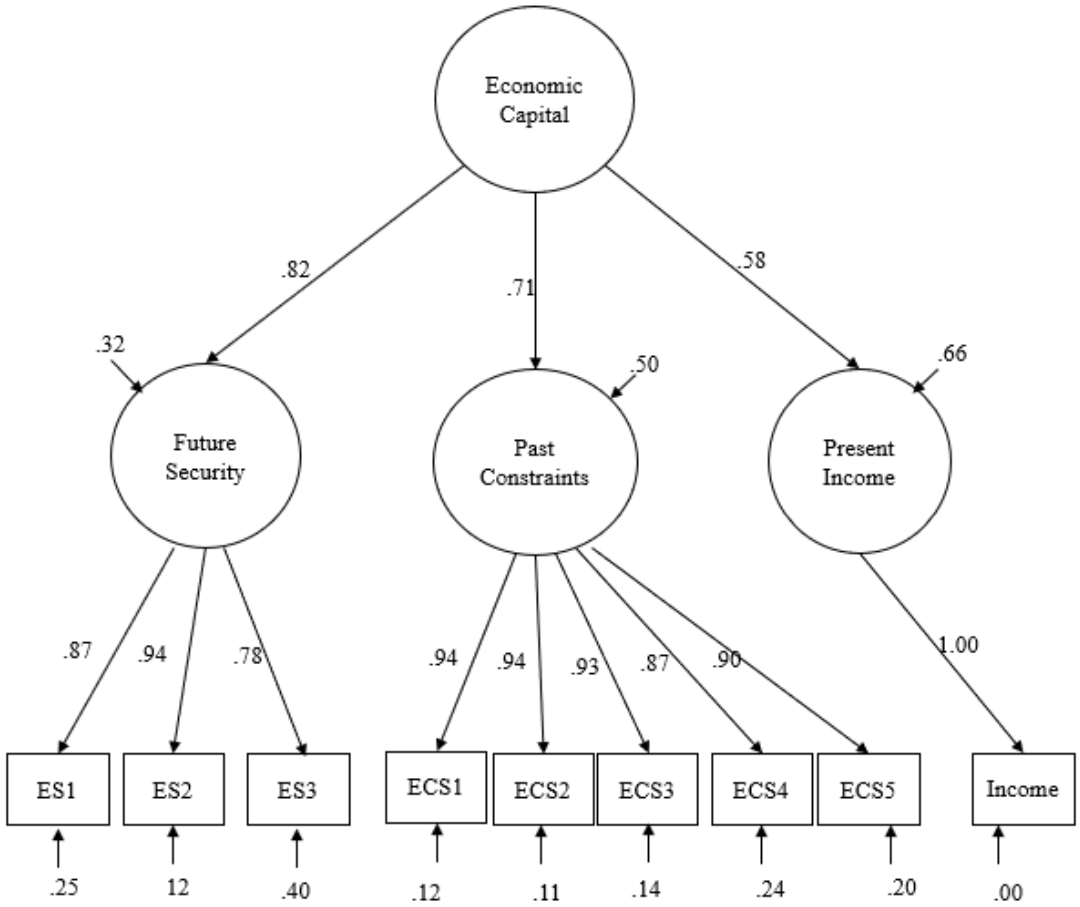
Model Fit Comparison for Economic Capital

		χ^2	*df*	χ^2/df	CFI	RMSEA	RMSEA 90% CI	SRMR	AIC	Adjusted BIC
1	One-factor model	658.79	27	24.40	.89	.07	[.18, .20]	.12	19155	19178
2	Three- factor model	54.47	25	2.18	0.995	0.04	[.03, .06]	0.03	18305	18394
3	Three-factor with a higher order factor model	54.47	25	2.18	0.995	0.04	[.03, .06]	0.03	18305	18394

Note. N = 629. Absolute indexes: χ^2/df = normed chi-square. Relative indexes: CFI = comparative fit index. RMSEA = root-mean-square error of approximation. CFI = robust comparative fit index; RMSEA = robust root-mean; square error of approximation; SRMR = standardized root mean square residual; AIC = The Akaike information criterion; BIC = Bayesian Information Criteria.Criteria used to determine model fit: CFI ≥ .95, (χ^2/df) ≤ 5, RMSEA ≤ .08, and SRMR ≤ .06 (Hu & Bentler, 1999). Lower values of AIC and BIC suggest better model parsimony.

Figure 4

Graphical Representation of a Higher-order Factor with a Three-factor Model for Economic Capital



Note. N = 629. Circles represent latent factors, and squares represent manifested variables. Single-headed arrows index factor loading. Single-headed arrows index factor loading and numbers next to each manifested variable represent error variance associated with each item. Model fit indices: CFI = .99, χ^2/df = 2.87, RMSEA = .06, SRMR = .03, and 90% CI = [.04, .07].

Measurement Invariance Testing

To further assess whether the structure of economic capital is comparable across different contexts, I conducted measurement invariance tests across gender (man or woman) and race (White or BIPOC: Black, Indigenous, and People of Color). I tested for four types of

increasingly strict measurement invariance using robust maximum likelihood estimate: configural (model with no constraints), metric (model with equality constraints imposed on the factor loadings), scalar (model with equality constraints additionally imposed on the intercepts), and strict (model with equal amount variance additionally imposed). Subsequently, I conducted model comparisons, and the results are displayed in Table 7. The configural invariance model demonstrated good fit indices. Model comparisons for the subsequent metric, scalar, and strict models indicated no significant differences. This provides initial support for the structural validity of economic capital.

Table 7

Economic Capital: Fit Indices for Measurement Invariance Testing

Invariance Type	Scaled χ^2	Scaling Factor	*df*	CFI	RMSEA	RMSEA 90% CI	SRMR	$\Delta\chi^2$	Δdf	ΔCFI
Gender (n = 625)										
Configural	87.79	1.29	50	0.993	0.06	[.04, .06]	0.03			
Metric	94.09	1.24	58	0.993	0.05	[.03, .07]	0.04	2.98	8	0.000
Scalar	104.80	1.23	63	0.992	0.05	[.03, .07]	0.04	10.92	5	-0.001
Strict	101.13	1.37	71	0.994	0.04	[.02, .06]	0.04	3.76	8	0.002
Race (n = 549)										
Configural	85.02	1.27	50	0.994	0.05	[.03, .06]	0.04			
Metric	97.55	1.21	58	0.993	0.05	[.03, .06]	0.04	12.07	8	-0.001
Scalar	102.11	1.20	63	0.993	0.05	[.03, .06]	0.04	4.34	5	0.000
Strict	114.06	1.47	71	0.992	0.05	[.03, .06]	0.04	12.55	8	-0.001
Education (n = 482)										
Configural	121.10	1.29	75	0.988	0.05	[.04, .08]	0.04			
Metric	139.52	1.21	91	0.987	0.06	[.04, .07]	0.06	14.88	16	-0.001
Scalar	147.08	1.20	101	0.988	0.03	[.04, .07]	0.06	7.27	10	0.001
Strict	158.25	1.26	117	0.989	0.05	[.03, .06]	0.62	13.87	16	0.001

Note. Scaling Factor = Yuan-Bentler scaling correction factor; The estimator used for the scaling factor was robust maximum likelihood estimation. Due to small sample sizes, "*others*" were excluded from gender analysis and only male and female were included. Race was grouped into two: white and BIPOC (Black, Indigenous, and People of Color). Education was compared across three groups: two-year degree, four-year degree, and master's degree. High school and doctoral degrees were excluded due to the small sample size. CFI = robust comparative fit index; RMSEA = robust root-mean; square error of approximation; SRMR = standardized root mean square residual.

Additional Validity Testing

To test whether economic capital is positively related to well-being, I first conducted a linear regression analysis. The result indicated that economic capital was a significant and strong predictive variable for well-being ($\beta = .54$, $p < .001$). economic capital explained 29% of the variance in well-being. Next, I conducted a series of hierarchical regression analyses to test incremental aspects of economic capital in explaining well-being above and beyond income, past economic constraints, and future economic security. I conducted a two-step hierarchical regression analysis, and the results are presented in Table 8. First, I entered income as a predictor in the first step. The regression coefficient was significant ($\beta = .35$, $p = .001$), indicating that income is a significant predictor for well-being. This model explained 12% of the variance ($R^2 = .12$) in well-being. In the second step, I added economic capital as another predictor of well-being. The model explained 18% more variance in well-being, and it was a significant increase ($\Delta R^2 = .18$, $p < .001$). In another hierarchical regression analysis, I entered past constraints as a predictor in the first step and added economic capital in the second step. The result indicated that past constraints were a significant and strong predictive variable for well-being ($\beta = .41$, $p < .001$) in the step, and adding economic capital in the second step significantly explained 18% more variance in well-being ($\Delta R^2 = .13$, $p < .001$). Finally, in the other hierarchical regression analysis, I entered future economic security as a predictor in the first step and added economic capital in the second step. The result indicated that future economic security was a significant and strong predictive variable for well-being ($\beta = .54$, $p < .001$) in the first step, and adding economic capital in the second step significantly explained 3% more variance in well-being ($\Delta R^2 = .03$, $p < .001$).

Table 8

Hierarchical Regression: Economic Capital Predicting Well-being Above and Beyond Income

	Well-being	β		R^2	ΔR^2
Step 1	Present income	.35	***	.12	
Step 2	Present income	.16	**		
	+ Economic capital	.66	***	.30	.18***

Note. N = 629. ** p=.008. *** p < .001.

Hierarchical Regression: Economic Capital Predicting Well-being Above and Beyond Lifetime Economic Constraints

	Well-being	β		R^2	ΔR^2
Step 1	Past economic constraints	.42	***	.17	
Step 2	Past economic constraints	.07			
	+ Economic capital	.59	***	.30	.13***

Note. N = 629. *** p < .001.

Hierarchical Regression: Economic Capital Predicting Well-being Above and Beyond Economic Security

	Well-being	β		R^2	ΔR^2
Step 1	Future economic security	.49	***	.29	
Step 2	Future economic security	.28	***		
	+ Economic capital	.30	***	.32	.03***

Note. N = 629. *** p < .001.

Results Summary

The structure of a higher-order factor with a three-factor model was tested using Confirmatory Factor Analysis. The results demonstrated good model fit indices (CFI = .99, Chi-square to degrees of freedom ratio (χ^2/df) = 2.87, RMSEA = .06, and SRMR = .03, and 90% CI = [.04, .07]), confirming initial validation of the structure of the economic capital in the present data. In addition, four steps of measurement invariance tests indicated insignificant differences between the configural model and the metrics model ($\Delta\chi2$ = 2.98 for gender and 12.07 for race), metrics model and scalar model ($\Delta\chi2$ = 10.92 for gender and 4.34 for race), and scalar model and strict model ($\Delta\chi2$ = 3.76 for gender and 12.55 for race), confirming the structure of economic capital is stable across different contexts (i.e., gender and race). Finally, a series of hierarchical regression analyses revealed that economic capital explains variance in well-being more comprehensively above and beyond income, past economic constraint, and future economic security. From all these results, I determined to utilize economic capital to identify economically marginalized (EM) workers for the following hypotheses. Economic capital scores below three were categorized as EM workers, and the total of the EM worker sample was n = 176. Scores of economic capital from three to seven were included in the non-EM workers and the total of the non-EM worker sample was n = 453.

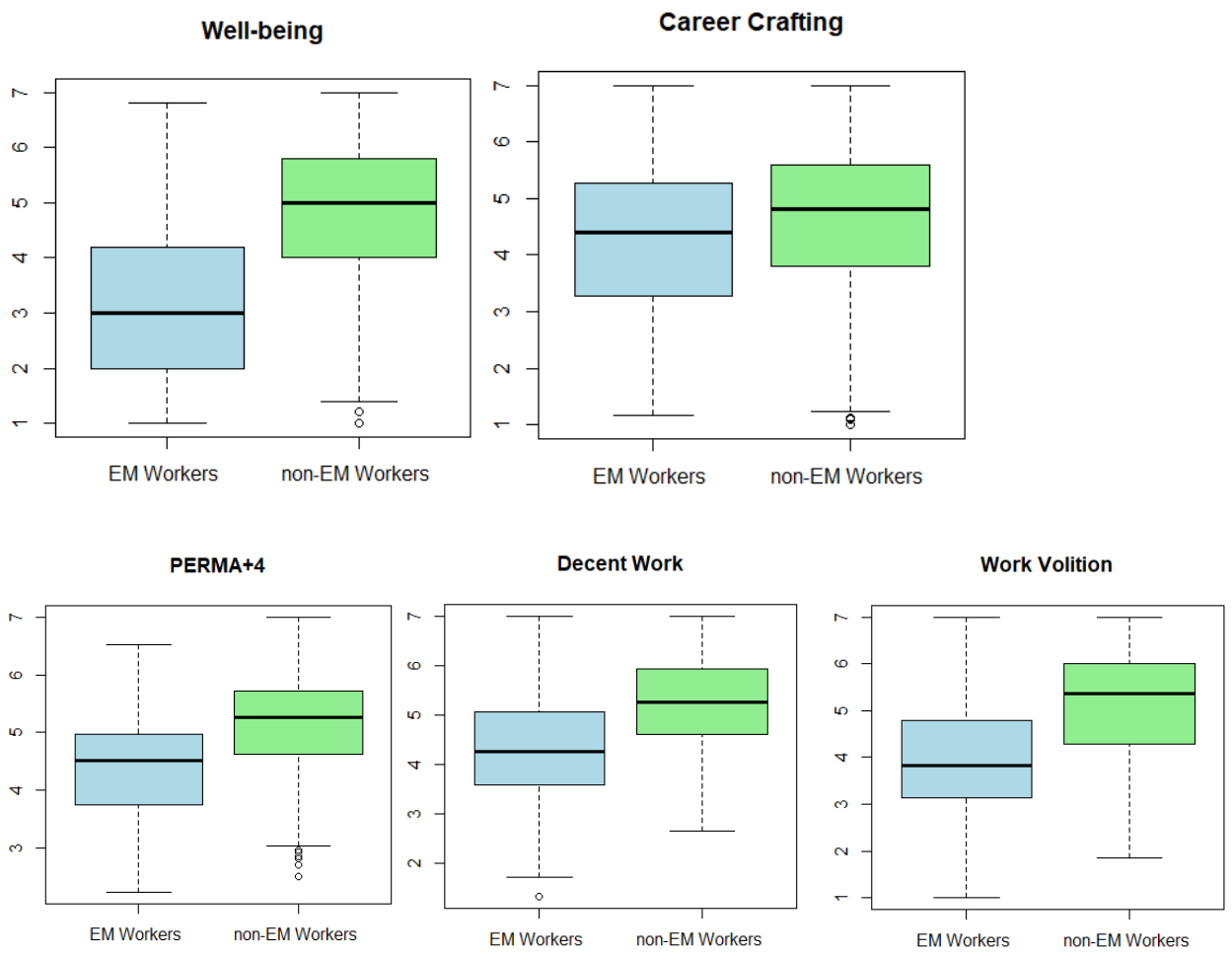
Two Group Comparisons

Figure 5 visualizes the mean differences between the groups. To test the significance of the group differences in study variables that include well-being, career crafting, decent work, PERMA+4, and work volition between economically marginalized (EM) and non-EM workers, t-tests were intended. Before conducting these t-tests, Levene's tests were performed to assess whether variances were equal across EM and non-EM groups. The results revealed insignificant p-values for well-being (p = .136), career crafting (p = .369), PERMA+4 (p = .084), and work

volition (p = .187), indicating equal variances between the two groups. However, the assumption of equal variance was violated for decent work, as indicated by a significant p-value (p = .004). Consequently, I conducted Welch's two-sample t-tests, which are recommended when dealing with unequal variances and possibly unequal sample sizes, providing more reliable results under these conditions (Derrick et al., 2016).

Figure 5

Visual Representation of the Mean Differences between EM and non-EM Workers



The t-test results, means, and standard deviations are presented in Table 9. These results confirm that economically marginalized (EM) workers reported significantly lower mean scores than non-EM workers in all study variables, including well-being (M = 3.23, SD = 1.46 for EM

workers; *M* = 4.77, *SD* = 1.36 for non-EM workers), career crafting (*M* = 4.25, *SD* = 1.32 for EM workers; *M* = 4.65, *SD* = 1.29 for non-EM workers), decent work (*M* = 4.29, *SD* = 1.09 for EM workers; *M* = 5.19, *SD* = 0.91 for non-EM workers), PERMA+4 (*M* = 4.70, *SD* = 0.90 for EM workers; *M* = 5.26, *SD* = 0.82 for non-EM workers), and work volition (*M* = 3.92, *SD* = 1.19 for EM workers; *M* = 5.13, *SD* = 0.91 for non-EM workers). Each of the adjusted p-values, having controlled for family-wise error, confirmed significant mean differences in well-being, career crafting, decent work, PERMA+4, and work volition between EM and non-EM workers.

Table 9

T-tests: Mean comparisons between EM and non-EM Workers

Variables	Group	n	*M*	*SD*	95% CI	*t* (*df*)	*Adjusted p*	Cohen's *d*
Well-being	EM	176	3.23	1.46				
	Non-EM	453	4.77	1.36	[-1.79, -1.29]	-12.09 (300.18)	***	-1.11
Career crafting	EM	176	4.25	1.32				
	Non-EM	453	4.65	1.29	[-0.63, -0.17]	-3.37 (307.73)	.004	-.31
Decent work	EM	176	4.29	1.09				
	Non-EM	453	5.19	0.91	[-1.08, - 0.71]	-9.68 (275.13)	***	-.93
PERMA+4	EM	176	4.70	0.90				
	Non-EM	453	5.26	0.82	[-0.94, -0.63]	-10.07 (293.5)	***	-.66
Work volition	EM	176	3.92	1.19				
	Non-EM	453	5.13	0.91	[-1.42, -1.01]	-11.76 (294.52)	***	-1.22

Note. Welch's two-sample t-tests. Adjusted *p*: *p*-values after controlling family-wise error using the Bonferroni method. EM = economically marginalized workers.

*** $p < .001$.

The effect sizes of the group differences on well-being, PERMA+4, and work volition were large (Cohen's *d* = -1.11, -0.93, and -1.22, respectively), while the effect size on

PERMA+4 was moderate to large (Cohen's d = -0.61), and the effect size on career crafting was small to moderate (Cohen's d = -0.31; Cohen, 1977; 1988), Together, the results from the t-tests and Cohen's d indicate that the differences in well-being, career crafting, decent work, PERMA+4, and work volition were statistically significant and practically meaningful on a relatively moderate to large scale.

Hypothesis 2

Multiple regression analysis was performed to examine whether economic capital, career crafting, and decent work are significant predictive variables for well-being (Hypothesis 2). The regression analyses results are presented in Table 10. First, for the entire sample (N = 629), the regression coefficients indicated that economic capital ($\beta = .40$, $p < .001$), decent work ($\beta = .20$, $p < .001$), and career crafting ($\beta = .18$, $p < .001$) were significant predictors, capturing 37% of the variance in well-being. Subsequently, I conducted a multiple regression analysis for economically marginalized (EM) workers (n = 176). Similar to the entire sample, economic capital ($\beta = .15$, $p < .05$), decent work ($\beta = .22$, $p < .01$), and career crafting ($\beta = .20$, $p < .01$) emerged as significant predictors, explaining 16% of the variance in well-being. Finally, the analysis in non-EM workers (n = 453) revealed a consistent pattern: economic capital ($\beta = .28$, $p < .001$), decent work ($\beta = .20$, $p < .001$), and career crafting ($\beta = .19$, $p < .001$) were significant predictors, with the model accounting for 23% of the variance in well-being. In summary, multiple regression analyses across different groups supported that economic capital, decent work, and career crafting are significant predictors of well-being, explaining a substantial portion of the variance in well-being across the entire sample, EM workers, and non-EM workers.

Table 10

Multiple Regression Analysis Results: Economic Capital, Decent Work, and Career Crafting as Predictors for Well-being

(1) All Sample (*N* = 629)

	β	*SE*	*p*	*Adjusted* R^2
Outcome: Well-being				.37
Economic Capital	.40	.03	***	
Decent Work	.20	.04	***	
Career Crafting	.18	.03	***	

Note. *** $p < .001$.

(2) EM Worker Group (n = 176)

EM Workers (n =176)	β	*SE*	*p*	*Adjusted* R^2
Outcome: Well-being				.16
Economic Capital	.15	.07	*	
Decent Work	.22	.08	**	
Career Crafting	.20	.08	**	

Note. EM = economically marginalized. ** $p < .01$. * $p < .05$.

(3) Non-EM Worker Group (n = 453)

Non-EM Workers (n =453)	β	*SE*	*p*	*Adjusted* R^2
Outcome: Well-being				.23
Economic Capital	.28	.04	***	
Decent Work	.19	.05	***	
Career Crafting	.20	.04	***	

*** $p < .001$.

Hypothesis 3

To test if career crafting positively contributes to the variance in decent work, which in turn contributes to variance in well-being, while controlling work volition, I performed the quasi-Bayesian Monte Carlo simulation, which allows estimates for direct, indirect, and total effects

(see Tingley et al., 2014). In the first step, a regression model was established, including career crafting and work volition as predictors and decent work as an outcome. Then the second regression model was established, including career crafting, work volition, and decent work as predictors and well-being as an outcome. Finally, the quasi-Bayesian Monte Carlo simulation mediation analysis was performed 1000 times with different bootstrap samples to estimate the distribution of the indirect effect and calculate confidence intervals.

Table 11 presents the direct effect of career crafting on well-being, the indirect effect of career crafting on well-being through decent work, and the total effect. In the EM group, it was found that there was a significant direct effect, with an estimated effect size of .19, 95% CI [0.03, 0.34], p = .028. A significant indirect effect via decent work was also found with an estimated effect size of .04, 95% CI [-0.00, 0.10], p = .040. The total effect of the model was found to be significant, with an estimated effect size of .23, 95% CI [-0.08, 0.38], p < .001. The proportion of the mediation effect suggested that decent work explained 19% of the relationship between career crafting and well-being in the EM group. When sub-factors of decent work were further analyzed, Value was the only significant contributor in the relationship between career crafting and well-being (estimated effect size = .10, 95% CI [0.04, 0.17], p < .001) whereas the indirect effect of career crafting on well-being via the other four factors (safety, healthcare, compensation, and rest) was insignificant. This result partially supports hypothesis 3.

In the non-EM workers, the overall relationship between career crafting, decent work, and well-being was similar. There was a significant direct effect, β = .18, 95% CI [0.07, 0.35], p < .001. The indirect effect of career crafting on well-being through decent work was significant (β = .05, 95% CI [0.02, 0.08], p = .002.), and the direct effect of career crafting on well-being was also significant (estimate = .20, 95% CI [0.12, 0.30], p < .001). Decent work explained 18%

of the relationship between career crafting and well-being in the non-EM group. Further analysis with sub-factors of decent work showed that compensation (estimate = .03, 95% CI [0.01, 0.05], p = .004) and value (estimated effect size = estimated effect size = .06, 95% CI [0.03, 0.10], p < .001) were variables that significantly mediated the relationship between career crafting and well-being.

Table 11

Indirect Effects of Decent Work Between Career Crafting and Well-being

Mediating Variable	Indirect Effect			Direct Effect			Total Effect			Prop Med Effect		
	Est.	[95% CI]		Est.	[95% CI]		Est.	[95% CI]		Est.	[95% CI]	
EM (n = 176)												
Decent Work	.04	[.00, .10]	*	.19	[.03, .34]	*	.23	[.08, .38]	***	.19	[.01, .65]	*
Safety	.01	[-.02, .04]		.22	[.06, .38]	**	.23	[.07, .38]	***	.02	[-.10, .24]	
Healthcare	.00	[-.02, .04]		.22	[.06, .38]	**	.23	[.07, .39]	**	.02	[-.18, .22]	
Compensation	.15	[-0., .05]		.21	[.06, .37]	***	.23	[.07, .39]	**	.05	[-.03, .33]	
Rest	.00	[-.02,.02]		.23	[.08, .38]	**	.23	[.07, .38]	**	.00	[-.11, .09]	
Value	.10	[.04,.17]	***	.13	[-.02, .29]		.23	[.07, .39]	***	.43	[.17, 1.36]	***
Non-EM (n = 453)												
Decent Work	.05	[.02, .08]	***	.20	[.12, .30]	***	.25	[.16, .34]	***	.18	[.07, .35]	***
Safety	.00	[-.00, .01]		.25	[.16, .34]	***	.25	[.16, .34]	***	.00	[-.03, .05]	
Healthcare	.01	[-.01, .04]		.24	[.15, .32]	***	.25	[.16, .33]	***	.05	[-.04, .17]	
Compensation	.03	[.01, .05]	**	.22	[.13, .32]	***	.25	[.16, .34]	***	.12	[.04, .23]	**
Rest	.00	[-.01, .01]		.25	[.16, .34]	***	.25	[.16, .34]	***	.00	[-.05, .02]	
Value	.06	[.03, .10]	***	.19	[.09, .28]	***	.25	[.16, .34]	***	.24	[.10, .44]	***

Note. 95% CI = quasi-bayesian confidence intervals. The number of Bootstrap simulations = 1000. *** $p < .001$. ** $p < .01$. * $p < .05$.

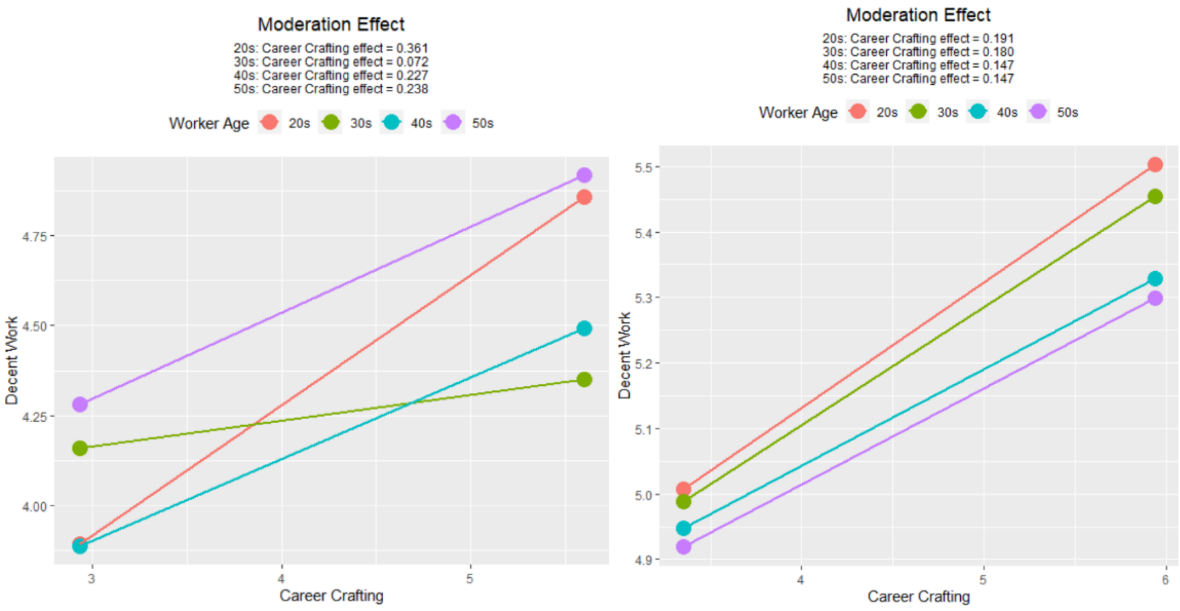
Est. = coefficient estimate. Prop Med Effect = proportion of the mediation effect. Work volition was included as a covariate.

Hypothesis 4

Moderation analysis was performed to examine the role of worker age in the relationship between career crafting and decent work in EM workers. The workers' age was factorized as 20s, 30s, 40s, and 50s+, and an interaction term (age * career crafting) was centered and included as a predictor in regression analysis. No significant age effects were found in both groups, and thus hypothesis 4 was not supported. However, when using the 20s as a reference group, although statistically insignificant (p = .059), visual inspection (see Figure 7) of the 30s age group suggests that when workers were in their 30s, the same level of career crafting may be slightly less associated with decent work compared to workers of other age groups. In the non-EM group, neither statistical age effect nor different patterns from visual inspection were found on the relationship between career crafting and decent work. Figure 6 presents a visual inspection of the moderation analysis results in the EM and non-EM groups.

Figure 6

Visual Graphs of the Relationship between Career Crafting and Decent Work by Age



EM group Non-EM group

Note. n = 45, 75, 41, 18 for the 20s, 30s, 40s, and 50s in the EM group. n= 80, 206, 103, and 48, for the 20s, 30s, 40s, and 50s for the Non-EM group.

Hypothesis 5

Next, I conducted another quasi-Bayesian Monte Carlo simulation to assess the mediating role of PERMA+4 in the relationship between career crafting and well-being. The first regression model was established with career crafting as a predictor and PERMA+4 as the outcome. The second regression model was established with career crafting and PERMA+4 as predictors and well-being as the outcome. This mediation analysis was conducted 1,000 times with different bootstrap samples to estimate the distribution of the indirect effect and calculate confidence intervals.

Table 12 presents the indirect effect of career crafting through PERMA+4. The results indicated that the indirect effect of career crafting on well-being through PERMA+4 was significant in the EM group (estimated effect size = .27, 95% CI [.17, .39], $p < .001$). The direct relationship between career crafting and well-being became insignificant, suggesting a full mediating role of PERMA+4. The proportion of the relationship between career crafting and well-being captured by PERMA+4 was 78%. Similarly, in the non-EM group, the indirect effect of career crafting on well-being through PERMA+4 was significant and large (estimated effect size = .35, 95% CI [0.27, 0.44], $p < .001$), and the direct effect of career crafting on well-being was insignificant. PERMA+4 captured 105% of the relationship between career crafting and well-being in the non-EM group[2]. All these results support hypothesis 4.

[2] A proportion greater than 1 would be due to the negative directions of the direct effect of career crafting on well-being, although small and insignificant.

Table 12

Indirect Effects of PERMA+4 Between Career Crafting and Well-being

Mediating Variable	Indirect Effect		Direct Effect		Total Effect		Prop Med Effect	
	Est.	[95% CI]	Est.	[95% CI]	Est.	[95% CI]	Est.	[95% CI]
EM (n = 176)								
PERMA+4	.27	[.17, .39] ***	.07	[-.10, .25]	.35	[.18, .49] ***	.78	[.45, 1.52] ***
Positive emotion	.19	[.10, .30] ***	.15	[-.02, .31]	.34	[.18, .49] ***	.57	[.27, 1.11] ***
Engagement	.05	[-.02, .13]	.29	[.11, .47] ***	.34	[.19, .50] ***	.16	[-.07, .46]
Relationships	.04	[-0.2, .11]	.30	[.12, .47] ***	.34	[.18, .49] ***	.12	[-.06, .39]
Meaning	.16	[.06,.26] ***	.18	[.00, .35] *	.34	[.18, .49] ***	.48	[.17, .98] ***
Accomplishment	.42	[-.02, .11]	.30	[.12, .47] ***	.34	[.18, .49] ***	.12	[-.05, .39]
Health	.04	[-.00,.09]	.30	[.15, .45] ***	.34	[.18, .50] ***	.11	[-.00, .29]
Environment	.04	[-.00, .09]	.31	[.02, .05] ***	.34	[.18, .49] ***	.10	[-.01, .28]
Economic security	.03	[-.01,.06]	.33	[.18, .49] ***	.34	[.34, .19] ***	.03	[-.06, .15]
Growth mindset	.12	[.04, .20] **	.22	[.06, .40] *	.34	[.19, .48] ***	.34	[.10, .73] **
Non-EM (n = 453)								
PERMA+4	.35	[.27, .44] ***	-.02	[-.13, .09]	.33	[.24, .42] ***	1.05	[.76, 1.52] ***
Positive emotion	.27	[.19, .34] ***	.07	[-.04, .18]	.33	[.24, .43] ***	.80	[.54, 1.17] ***
Engagement	.04	[.01, .07] *	.30	[.20, .39] ***	.34	[.25, .43] ***	.11	[.03, .23] *
Relationships	.04	[-0.2, .11 ***	.30	[.12, .47] ***	.34	[.18, .49] ***	.12	[-.06, .39] ***
Meaning	.12	[.06,.26] ***	.21	[.00, .35] ***	.34	[.24, .43] ***	.36	[.18, .64] ***
Accomplishment	.14	[.09, .19] ***	.19	[.10, .29] ***	.33	[.24, .43] ***	.42	[.26, .63] ***
Health	.05	[.02, .09] ***	.28	[.18, .37] ***	.33	[.24, .42] ***	.15	[.07, .27] ***
Environment	.08	[.04, .11] ***	.33	[.24, .42] ***	.32	[.32, .49] ***	.19	[.10, .29] ***
Economic security	.09	[.05,.13] ***	.24	[.16, .33] ***	.33	[.24, .42] ***	.27	[.17, .40] ***
Growth mindset	.22	[.15, .28] ***	.12	[.01, .22 *	.33	[.24, .43] ***	.65	[.44, .95] ***

Note. 95% CI = quasi-bayesian confidence intervals. The number of bootstrap simulations = 1000.

*** $p < .001$. ** $p < .01$. * $p < .05$. Est. = coefficient estimate. Prop Med Effect = proportion of the mediation effect.

Further analysis of the nine building blocks of PERMA+4 indicated that positive emotions (estimated effect size = .19, 95% CI [0.10, 0.30], $p < .001$), positive meaning (estimated effect size = .16, 95% CI [0.06, 0.26], $p < .001$), and ggrowth mindset (estimated effect size = .12, 95% CI [0.04, 0.20], $p < .001$) mediated the relationship between career crafting and well-being in the EM group. Positive emotions, positive meaning, and growth mindset captured 57%, 48%, and 34% of the relationship between career crafting and well-being, respectively.

Hypothesis 6

Multi-group Path Analysis: Originally proposed Model

I analyzed the entire study model initially proposed. Variables included well-being, career crafting, decent work, PERMA+4, Age, and work volition. The initially obtained model fit, without the path from work volition to PERMA+4, was poor, so I added a path from work volition to PERMA+4 to improve the model fit. Then, to ensure the data was appropriate for multi-group path analysis, I first tested the structural invariance across the EM and non-EM groups to determine if the model structure and paths were comparable across the groups. I used a robust maximum likelihood estimate and established two models to compare: an unconstrained model and a constrained model. The unconstrained model allowed all coefficients to be freely estimated across groups whereas the constrained model imposes equality constraints across the groups. Career crafting and work volition were entered as exogenous variables and treated as random, introducing additional free parameters in the unconstrained model. In the constrained model, they were conditioned as fixed. The results of the model comparison are summarized in Table 13. However, there was a significant difference between the two models ($p = .018$), indicating that the structures of the models are incomparable. Therefore, statistical differences

between the two models were not tested while testing each group independently to examine the patterns of the paths.

Table 13

Originally Proposed Model: Structural Invariance Testing Results for EM/non-EM Groups

Model Type	Scaled χ2	Scaling factor	*df*	χ2/*df*	CFI	RMSEA	RMSEA 90% CI	SRMR	Δχ2	*Δdf*	AIC
Constrained	17.5	0.99	7	2.5	0.995	0.098	[.03, .11]	0.023			10223.9
Unconstrained	8.05	1.01	2	4.02	0.996	0.034	[.04, .10]	0.017	11.91*	4	8411.9

Note. N = 629. * *p* < .05

Constrained = a model that imposes equality constraints across the groups. Unconstrained = a model that allows free parameters for endogenous variables.

In examining the regression coefficients, I used an unconstrained model because it is more parsimonious than the constrained model with lower AIC. Table 14 presents regression coefficients for each of the EM and non-EM groups. Note that because of the violation of the structural invariance, the coefficient differences between the two models were not statistically compared.

Table 14

Originally Proposed Path Model: Path Coefficients for EM and Non-EM Groups

		EM (n = 176)		Non-EM (n = 453)	
		β	*SE*	β	*SE*
Career Crafting	→ Decent Work	.20 ***	.06	.17 ***	.03
Work Volition	→ Decent Work	.31 ***	.06	.41 ***	.03
Career Crafting *Age	→ Decent Work	-.03	.03	-.03	.03
Career Crafting	→ PERMA+4	.28 ***	.10	.34 ***	.02
Work Volition	→ PERMA+4	.36 ***	.05	.33 ***	.03
Decent Work	→ Well-being	.11	.12	.06	.09
Career crafting	→ Well-being	.06	.10	-.01	.06
PERMA+4	→ Well-being	.64 ***	.17	.81 ***	.10

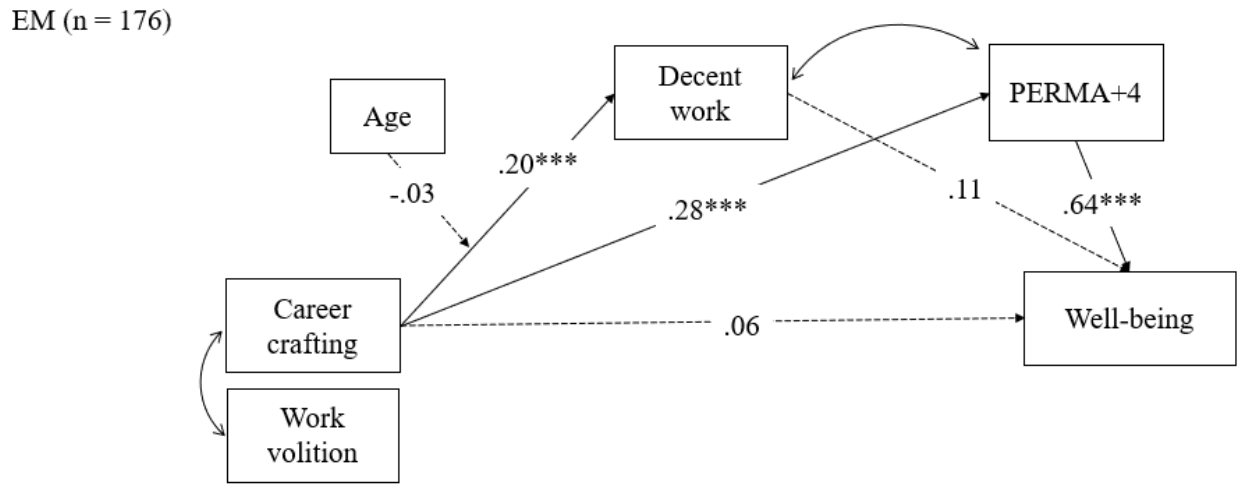
Note. β = Standardized Regression Weights; SE= Standard error.

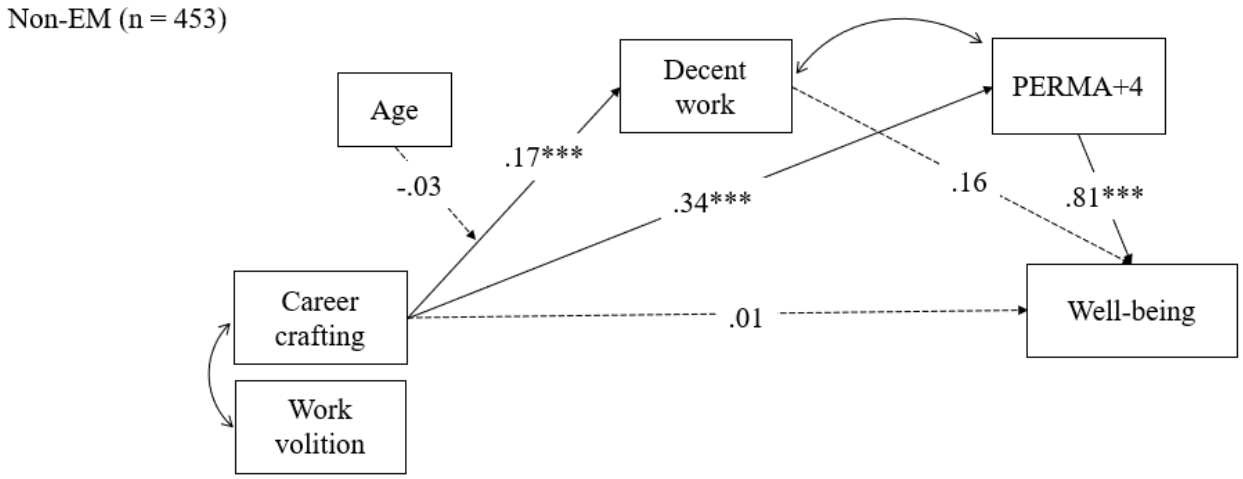
*** $p < .001$.

The model presented good fit indices. EM and non-EM groups showed a similar pattern. In the EM group, the paths from career crafting to decent work (β = .20, $p < .001$), work volition to decent work (β = .31, $p < .001$), career crafting to PERMA+4 (β = .28, $p < .001$), work volition to PERMA+4 (β = .36, $p < .001$), and PERMA+4 to well-being (β = .64, $p < .001$) demonstrated significant relationships. Similarly, in the non-EM group, the paths from career crafting to decent work (β = .17, $p < .001$), work volition to decent work (β = .41, $p < .001$), career crafting to PERMA+4 (β = .34, $p < .001$), work volition to PERMA+4 (β = .33, $p < .001$), and PERMA+4 to well-being (β = .81, $p < .001$) demonstrated significant relationships. Figure 7 presents a graphic representation of this proposed model.

Figure 7

Originally Proposed Model: Graphic Representation of Each Group's Path Analysis





Note. Solid lines indicate significant paths. Dotted lines indicate insignificant paths. The arrows with one direction are regression paths. The values are regression coefficients. Bi-directional arrows indicate covariates.

*** $p < .001$.

Multi-group Path Analysis: Improved Model

An improved final model was explored by removing insignificant paths (i.e., Age, career crafting to well-being, and decent work to well-being) and adding a path from decent work to PERMA+4. The structure invariance test was run using constrained and unconstrained models. The result is presented in Table 15. The result confirmed structural invariance across EM and non-EM groups, confirming that the direct statistical comparison of the coefficients between the EM and non-EM groups in the multi-group analysis is appropriate.

Table 1 5

Improved Final Model: Structural Invariance Testing Results

Structure Type	Scaled χ2	Scaling factor	*df*	χ2/*df*	CFI	RMSEA	RMSEA 90% CI	SRMR	Δχ2	Δ*df*	AIC
Constrained	9.016	0.173	6	1.50	0.998	0.04	[.00, .09]	0.02			4390
Unconstrained	9.016	1.14	6	1.50	0.998	0.04	[.00, .09]	0.02	0	0	8405

Note. N = 629. Constrained = a model that imposes equality constraints across the groups. Unconstrained = a model that allows free parameters for endogenous variables. A smaller AIC indicates a more parsimonious model.

Table 16 presents the results of the comparison of path coefficients between the EM and non-EM groups. I used the constrained model with a robust maximum likelihood estimate in examining the improved final model. Among the paths between the main study variables, significant coefficient differences were found in the relationships between decent work and PERMA+4 ($z = -11.30$, $p < .001$) and PERMA+4 and well-being ($z = -5.00$, $p < .001$) whereas the relationships between career crafting and decent work and career crafting and PERMA+4 did not differ between the EM and non-EM groups. Figure 8 presents a graphic representation of this improved model.

Table 16

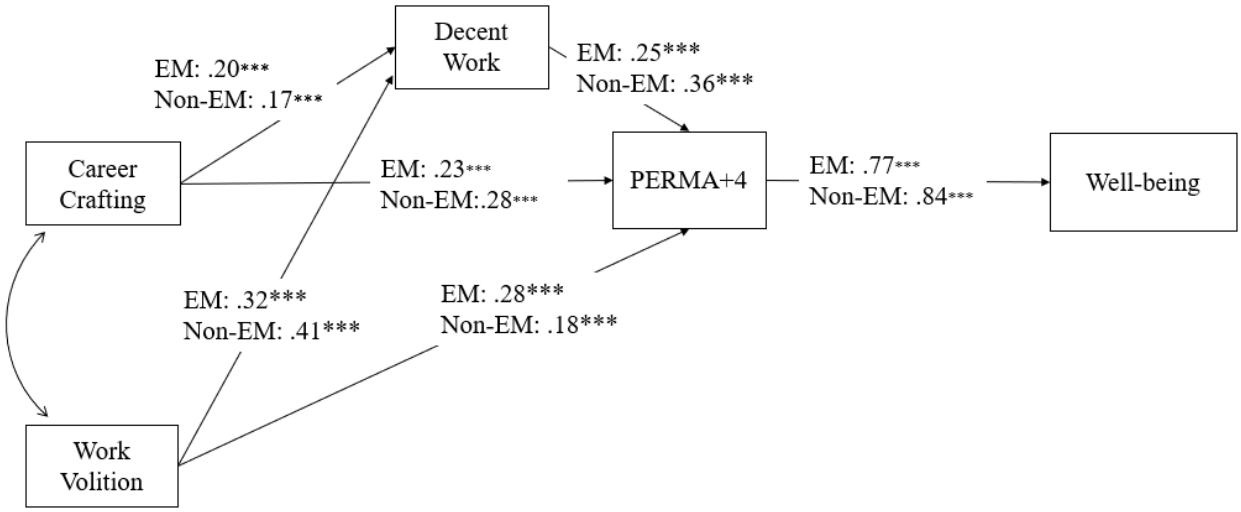
Improved Final Model: Comparison of Path Coefficients between EM and non-EM Groups

	EM (n = 179)		Non-EM (n = 453)		EM vs non-EM
	β	*SE*	β	*SE*	z-score
Career Crafting → Decent Work	.20 ***	.06	.17 ***	.03	1.15
Work Volition → Decent Work	.32 ***	.06	.41 ***	.03	-3.00 **
Career Crafting → PERMA+4	.23 ***	.39	.28 ***	.02	-0.14
Work Volition → PERMA+4	.28 ***	.04	.18 ***	.03	5.16 ***
Decent Work → PERMA+4	.25 ***	.04	.36 ***	.03	-11.30 ***
PERMA+4 → Well-being	.77 ***	.11	.81 ***	.10	-5.00 ***

Note. β = standardized regression weights; SE = standard error. EM = economically marginalized workers. *** $p < .001$. ** $p < .01$

Figure 8

Improved Final Model: Graphic Representation of Multi-group Path Analysis



Note. N = 629. EM = economically marginalized workers (n = 176). Non-EM (n = 453). The values represent standardized regression weights.

*** $p < .001$.

Chapter 5: Discussion

The present study investigated the relationships between career crafting, work volition, decent work, PERMA+4, and well-being in economically marginalized (EM) workers. Well-being is the most frequently studied construct in positive psychology (Ackerman et al., 2018; Kim et al., 2018), yet the implications in the context of economic marginalization have been understudied even though approximately one-third of the U.S. population are economically marginalized workers. Moreover, while decent work is seen as central to well-being (ILO, 2012),

its relationship to PERMA+4, the building blocks of well-being, has not yet been addressed in any study.

Defining Economic Marginalization using Economic Capital

The present study first investigated how best to describe economic marginalization and utilized past economic constraints and future economic security in addition to income to define economic resources, named economic capital. Correlation results indicated that all three of these were strongly correlated to one another—past economic constraints were significantly correlated with current income ($r = .40$, $p < .001$) level, and future economic security ($r = .56$, $p < .001$). Blustein and colleagues (2001) point out that an individual's current income reflects not just their personal traits or abilities, but also their historical access to financial resources. McWhirter & McWha-Hermann (2021) also noted that economic scarcity across one's lifetime should be approached as one experiencing marginalization because those who have faced financial struggles in the past are more likely to have lower incomes and lack savings for the future. This study's results support this perspective.

Economic capital, the defined concept that uses past economic constraints and future economic security, statistically demonstrated that it is a more comprehensive approach to understanding economic marginalization status than just income alone in predicting well-being. Confirmatory factor analysis confirmed that the three-factor model provides good fit indices. Also, measurement invariance tests were conducted to ensure the data is reliable when using more constrained models (i.e., configural, metric, scalar, and strict models). The results demonstrated indifferences between the model and the more constrained model, ensuring initial construct validity in this data. The hierarchical regression analyses confirmed that economic

capital is a significant and robust predictive variable for well-being ($\beta = .40$, $p < .001$), and explained 18% more variance in well-being, above and beyond income ($\Delta R^2 = .18$, $p < .001$).

While income is a typical way to gauge one's economic status, Duffy et al. (2012) argue that the level of economic constraint in the past should not be ignored, revealing its negative relationships to decent work. Donaldson and Donaldson (2021) propose that economic security should be a path to well-being and report its positive relationships to well-being. This study's results supported their argument and further advance it by synthesizing all three (i.e., past economic constraints, income, and future economic security) into a concept of economic capital to describe one's economic resources comprehensively. Furthermore, the economic context has rarely been considered in studies of well-being within positive psychology, perhaps due to the measurement problem (e.g., Inglehart et al., 2008; Kahneman & Deaton, 2010). For instance, in Shin and Lyubomirsky's (2009) meta-analysis, while several moderators were studied, including age and participants' depression status, the economic context did not appear. Utilizing economic capital can be a way to measure economic context more comprehensively with less concern.

Further analysis of the group comparison revealed that the well-being of marginalized workers was significantly lower than that of non-EM workers. Additionally, decent work, career crafting, and PERMA+4 were also significantly lower among EM workers. This confirms that the impact of economic resources on positive psychological states cannot be ignored, nor can the challenge of establishing tangible and intangible career resources be overlooked. This underscores the need to pay more attention to the needs of these populations to adequately address economic inequality.

Economic Capital, Career Crafting, Decent Work, and Well-being

According to the conservation of resources theory (COR), lacking resources negatively impacts well-being (Hobfoll, 2002). The theory also argues that reserving alternative resources serves as a means to achieve goals and has a positive psychological impact (Hobfoll, 1988, 1989, 2011). When it comes to career context, the identified resources investigated in this study were career crafting, decent work, and economic capital. Economic capital represents a resource for fulfilling one's financial needs, while decent work promotes human dignity. Career crafting, on the other hand, is examined as a resource for career enactment.

When controlling economic capital, decent work was still a significant predictor for well-being. Career crafting was also a significant predictor of well-being while holding economic capital constant. Together, when all these three predictors were entered as predictors in multiple regression, each predictor showed significant and positive relationships with well-being, suggesting their unique contributions to well-being ($\beta = .40$, $p < .001$; $\beta = .20$, $p < .001$; $\beta = .18$, $p < .001$ for economic capital, decent work, and career crafting, respectively). These relationships appeared similarly for EM and non-EM workers. Although EM workers economically struggle, having work that taps into satisfying human dignity (decent work) and establishing career resources (career crafting) can help increase their well-being. It should be noted, however, that decent work and career crafting would not be able to replace the absence of economic resources in achieving well-being. However, the stronger relationship between well-being and decent work, as well as between well-being and career crafting compared to the relationship between economic capital and well-being among EM workers, emphasizes the importance of cultivating career resources beyond economic resources to enhance overall well-being. Conservation of Resource theory posits that resources are interconnected and influence one's perception of resource gain or loss as a total (Hobfoll, 1988, 1989, 2011; Hobfoll & Schumm, 2009). While

this result does not support the notion that acquired resource reserves can offset losses, it suggests that although inadequate financial resources may have a negative impact on well-being, perceiving work as decent and engaging in career crafting behaviors can facilitate the acquisition of other types of resources that can have a positive effect on overall well-being.

Findings on Decent Work

Participants experience greater well-being when their jobs provide psychological and physical safety, adequate compensation and healthcare and rest time, and match their personal or family values. As Blustein et al. (2001) and Duffy et al. (2012) note, decent work can be a robust resource that satisfies human dignity.

Because career crafting generates tangible career resources such as new networks and skills, these practical resources were thought to increase the opportunity to access decent work. Work volition was included as a controlling variable informed by its critical role in the prior studies of the psychology of working theory (Duffy et al., 2012). Collectively, it was hypothesized that accounting for work volition, engaging in career crafting relates to greater access to decent work, which, in turn, relates to greater well-being (hypothesis 3). In multiple regression analysis, career crafting and work volition were significant predictors for decent work (β =.20, $p < .001$ and β =.32, $p < .001$) among the EM sample, explaining 24% of the variance in decent work. This result is that having both confidence and taking action work better for accessing decent work than having one of these, confirming their vital contributions to accessing decent work. As the next step, by adding well-being, the direct and indirect effects of career crafting on well-being were tested from the mediation analysis. Both the indirect path through decent work and the direct effect of career crafting on well-being were significant. This suggests that while workers highly engaged in career crafting have greater access to decent work and

experience higher well-being, their improved well-being is partially attributable to having decent work but related to career crafting itself. Similarly, in the non-EM group, the indirect path of career crafting via decent work was significant. Among the five elements of decent work, value fully mediated the relationship between career crafting and well-being, but other components did not contribute to their relationship in EM workers. Value was also a significant mediator but partially so in the non-EM workers, and compensation was another mediator that explained the relationship between career crafting and well-being. The significance of value congruence between worker and work has been emphasized in the literature on person-organization fit (Cable & DeRue, 2002; Kristof-Brown, 1996). When employees' personal goals and values align with those of their organization, they experience a higher level of person-organization fit, and research has demonstrated that this is crucial for retaining talented employees (Van Vianen, 2000). Career crafting assists workers in finding work that aligns with their values regardless of their economic contexts, and such value congruence in work contributes to overall well-being.

The relationship between career crafting and the attainment of decent work did not vary across the worker's age. The moderation analysis of the present study indicates that among the EM group, workers in their 20s facially appeared a stronger correlation between career crafting and decent work than those in their 30s, but the difference was not statistically significant ($p = .059$). No significant difference was observed between workers in their 20s and those in their 40s and 50s. Similarly, in the non-EM group, the relationship between career crafting and decent work. According to the U.S. Bureau of Labor Statistics, people in their 20s and 30s are more likely to change jobs than people of other ages, suggesting that younger workers are more actively seeking opportunities to align their careers with personal goals and aspirations, including the pursuit of decent work. Individuals in their 30s may place a higher emphasis on

financial stability and family responsibilities. This focus on economic security might lead them to prioritize positions that offer higher salaries, potentially at the expense of neglecting other components of decent work, such as rest time, physical safety, or personal values. Nonetheless, a more detailed examination of the elements of decent work is necessary to fully understand how age modulates the relationship between career crafting and the perception of decent work.

While decent work is an explicit and tangible outcome variable of career crafting, PERMA+4 reflects rather implicit resources generated in the process of career crafting. The indirect effect of career crafting on well-being throughout PERMA+4 is investigated to unravel the implicit sources linking career crafting to well-being. The mediation analysis revealed a significant indirect effect of career crafting on well-being through PERMA+4, fully explaining the relationship and nullifying the direct link between career crafting and well-being.

Findings on Career Crafting and PERMA+4 for Well-being

Career crafting generates concrete career resources such as new networks and skills, but also guides workers towards a more meaningful career aligned with one's interests, values, and abilities. This suggests actions involve generating positive psychological elements, and PERMA+4 provided insights into those psychological elements.

This study found that PERMA+4 fully mediated the relationship between career crafting and well-being in both EM and non-EM workers. To be specific, among the elements of PERMA+4, positive emotions, meaning, and growth mindset linked the relationship between career crafting and well-being in EM workers whereas other elements of PERMA+4, including positive engagement, positive relationships, positive accomplishments, positive health, economic security, and a positive work environment did not explain the relationship between career crafting and well-being. PERMA+4 covers a wide spectrum of pathways to well-being, though

not exhaustively. As Donaldson and Donaldson (2022) note, multiple paths intertwine and contribute to a greater well-being experience. In the case of career crafting, three out of nine elements were involved and fully or partially linked the relationship to well-being in EM workers, suggesting that the process of developing career resources through engaging in career crafting involves multiple elements related to well-being, with positive feelings, meaning, and growth mindset standing out as three pivotal pathways to experiencing greater well-being in the EM context. In the context of non-EM, all nine paths of PERMA+4 were shown to explain the relationship between career crafting and well-being: positive emotions fully mediated the relationship whereas the other eight components played a partial role in explaining this relationship.

PERMA+4 provides practical insight when designing career development programs. Because career programs by design aim to advance one's professional goals, the accompanying psychological benefits are seldom recognized. It is worth being aware that engaging in career crafting not only relates to pursuing tangible career success (see Lee et al., 2021), but also enriching psychological experiences, thereby relating to greater well-being. Furthermore, while it remains a debate whether behavior or attitude comes first (e.g., self-perception theory, Bem, 1972), the pathway identified in this mediation model suggests that behaviors play a crucial role in broadening the psychological repertoire. However, future longitudinal studies may offer deeper insights into this relationship.

The study hypotheses collectively comprise a complex model, as shown in Figure 6. Thus, the model was analyzed by conducting a path analysis to explore if the model describes the complex phenomenon well. Findings support that career crafting, in tandem with work volition, is critical for improving access to decent work. Furthermore, engaging in career crafting

behaviors contributed to PERMA+4, which, in turn, contributed to greater well-being. These results confirm that incorporating career crafting and work volition into career program strategies constitutes a crucial method for seeking decent work and improving overall well-being.

When examining the strength of the relationships between career crafting and decent work, and between career crafting and PERMA+4, no significant differences were found between EM and non-EM workers, suggesting that the implication of career crafting in relation to decent work and PERMA+4 is consistent across different economic contexts.

Career Crafting and Positive Emotions in EM Workers. Positive emotions were the only full mediator to connect the relationship between career crafting and well-being in EM workers (estimate indirect effect = .19, $p < .001$), while positive meaning and mindset were also significant but partial mediators while others were insignificant. This suggests that career crafting is an important source of positive emotions. Career crafting is composed of actions to build essential career resources externally and internally in the domains of relationships, cognition, and abilities. When these resources are being accumulated, they may help increase confidence and self-esteem and reduce negative thoughts and feelings. Research also shows that adopting a proactive stance predicts a positive future self, which is also related to positive emotions (Cai et al., 2015; Goodall, 2015).

The conservation of resources theory posits that individuals strive to acquire and maintain various resources to avoid negative emotions like anxiety and depression (Hobfall 1988, 1989, 2002). These resources include tangible objects, psychological attributes, and energy levels, and their acquisition can offset losses. For instance, research indicates that individuals who experience gains are less affected by subsequent losses. Hence, while insufficient financial

resources may harm well-being, engaging in career crafting behaviors could help acquire other resources, potentially reducing negative emotions and thus, greater well-being.

However, research on positive emotions has identified the opposite direction. According to broaden and build theory (Fredrickson, 2001), positive affect predicts improvements in behavior, which, in turn, leads to increases in positive affect. Later, they suggest that the built resources strengthen the relationship between positive behavior and positive affect, showing an upward spiral dynamic (Fredrickson, 2013; Fredrickson & Joiner, 2017; Van Cappellen et al., 2017). While directions are mixed, one clear takeaway is that both the conservation of resources theory and the broaden and build theory emphasize the importance of acquiring and maintaining resources for well-being, and it is both career crafting and positive emotions that generate resources.

Career Crafting and Meaning in EM Workers. Purpose and meaning within career development is an expanding field that bridges counseling and vocational psychology with positive psychology, leading to promising practices in career counseling (Dik et al., 2015). The current study found that among the PERMA+4, meaning partially mediated the relationship between career crafting and well-being in EM workers.

Gaining resources is identified as an important source of meaning, which may enhance various aspects of life, as noted by Frankl (2006). Frankl argues that when faced with life's challenges, one can only respond by taking responsibility. This notion is particularly relevant in the context of career crafting, which involves taking responsibility for acquiring career resources. This process becomes even more crucial in uncertain and incomprehensible environments, such as those faced by marginalized workers. For EM workers, connecting with meaningful relationships and taking extra but meaningful tasks may be essential for meaning-making. A

study by Plomp et al. (2019) reported that temporary workers perceive their crafting efforts as less impactful and receive fewer benefits than permanent workers, which demotivates them from proactively changing their work boundaries. Discovering work-related meaning may require intentional cognitive processes, especially in EM workers' contexts. In a qualitative study by Wrzesniewski and Dutton (2001), a hospital cleaning staff member who exhibited higher job crafting found greater meaning in her work. This indicates that cognitive crafting can significantly enhance work experience, contributing to a greater sense of meaning in work. Career crafting encompasses three aspects, one of which is cognitive crafting. This domain is fundamental to career crafting, where individuals engage in reflective practices to discern their careers' significance, interpreting them as a means to find greater meaning in their lives (Lee et al., 2021). Accordingly, this study found that career crafting contributed to positive meaning, which in turn contributed to the well-being of EM workers.

Career Crafting and Growth Mindset in EM Workers. Growth mindset is characterized by an optimistic, future-oriented view of life, where challenges or setbacks are seen as opportunities for growth (Donaldson & Donaldson, 2020). The indirect effect of career crafting on well-being through a growth mindset was significant, although partial. This suggests that EM workers who are highly engaged in career crafting tend to exhibit a greater growth mindset. Research has indicated that individuals with a growth mindset typically opt for more demanding tasks that challenge their current abilities because they believe these tasks will provide opportunities for growth and meaningful challenges to extend their capabilities (Donaldson et al., 2020; Donaldson & Donaldson, 2021; Dweck, 2006, 2016). In the career context, those who engage in career crafting pursue challenges, aiming for future careers that are more meaningful and better aligned with their evolving interests, values, and abilities (Lee et al.,

2022). Although future studies will reveal more, it may be that career crafting and a growth mindset could establish a positive gain cycle (Hobfoll, 1989, 2011), whereby workers gain resources through career crafting, leading to increased positivity and optimism. This heightened positivity, in turn, may encourage more active engagement in career crafting.

Findings on Work Volition

Although work volition was not the primary focus of this study, its inclusion in this discussion is warranted due to the significant roles found in this study analysis. Work volition is a term that captures the extent to which individuals perceive themselves as having the freedom to choose their careers (Duffy et al., 2012). This study's analysis revealed that work volition is significantly and positively related to decent work ($\beta = .32$, $p < .001$) and PERMA+4 ($\beta = .28$, $p < .001$) in both EM and non-EM workers. These findings underscore the crucial role that autonomy and control over one's career decision play not only in promoting the acquisition of decent work but also in enhancing overall well-being, as encapsulated by the PERMA+4 model. This was consistent with prior research that revealed the importance of possessing work volition in various marginalized contexts (Masdonati & Fournier, 2015; Schoon & Polek, 2011; Vuolo et al., 2014).

A particularly interesting aspect of the findings is the differential impact of work volition on decent work and PERMA+4 between EM and non-EM workers. The link between work volition and decent work was significantly weaker among EM workers, yet its relationship with PERMA+4 was notably stronger. Work volition involves feelings and attitudes related to self-esteem, self-efficacy, and locus of control (Kim & Lee, 2020). With higher work volition, EM workers may still encounter more obstacles than non-EM workers in securing decent work

(Blustein, 2001). Conversely, these obstacles do not necessarily hinder their ability to engage in positive psychological states encapsulated by PERMA+4.

Theoretical Implications

This study carries important theoretical implications. First, the findings of this study offer advanced insights into the criticism by van Zyl et al. (2013) and Blustein (2006). van Zyl et al. (2023) highlighted the criticism that positive organizational psychology (POP) may be influenced by capitalist motives, potentially commodifying positivity and fostering individualistic and consumeristic values in organizations, which concerns the risk that the commercialization of positive practices might compromise the genuine well-being of individuals and organizations. In vocational psychology, Blustein (2006) raised concerns that research has traditionally centered on demographics with fewer career obstacles and a greater likelihood of finding meaning in their work, thus overlooking the experiences and challenges faced by marginalized individuals. The present study confirms that economically marginalized (EM) workers significantly lack resources for well-being, including career crafting, work volition, decent work, and PERMA+4, thereby underscoring the importance of understanding EM workers from a new perspective. Additionally, among the nine building blocks of pathways to well-being, EM workers focused particularly on positive emotions, meaning, and a positive mindset, whereas non-EM workers utilized all nine different paths. This finding supports the concern that more studies are needed for EM workers and highlights the need for a nuanced understanding of how the two groups achieve well-being. At the same time, it challenges the existing biases within vocational psychology, showing that the focus on meaning is not only beneficial for the privileged population but may be more salient for EM workers. Therefore,

more research related to meaning, positive emotions, and mindset is warranted among EM workers when discussing career development.

Secondly, this study enriches the career literature by examining the relationship between career and well-being, by highlighting the role of decent work and pathways of PERMA+4 in this relationship, with particularly meaningful findings on the role of positive emotions, meaning, and a positive mindset in explaining the relationships between career crafting and well-being in EM workers. This reinvigorates the debate surrounding self-perception theory (Bem, 1976), which posits that individuals infer their internal states from their behaviors. In a career setting, it's plausible that career crafting involves building relational, cognitive resources, and skills, which in turn foster positive affect. This positive affect may facilitate resource building, with these resources then moderating the relationship between career crafting and positive affect. Contrary to this, many studies propose that positive inner states precede positive behaviors. For instance, Fredrickson and Joiner (2017) suggest that positive emotions and behaviors interact in an upward spiral dynamic. They argue that positive affect leads to behavioral improvements, which then further enhance positive affect, particularly in the health context. Additionally, accumulated resources serve as moderators between behavior and positive affect. These two conflicting perspectives highlight the need for further evidence to elucidate the relationship fully. Although a clear direction has not been established, a bidirectional relationship is conceivable and whether positive career actions lead to positive psychological states or vice versa remains a topic for investigation.

Third, this study advances the literature on well-being theory. PERMA+4 represents a concerted effort to more fully understand the diverse pathways to well-being. Considering the scant research on the relationship between PERMA+4 and career development, our findings

offer a valuable link between positive psychology and vocational psychology. This integration provides new theoretical insights into well-being theory by examining the relationships between career behaviors and positive internal states, thereby enhancing our comprehension of how individuals' career actions and confidence can proactively shape their perspectives on their work and enhance the well-being of marginalized workers. Moreover, the exploration of well-being paths revealed intriguing differences between EM workers and non-EM workers, offering sophisticated avenues for future research.

Fourth, this study expands our understanding of the psychology of working theory (PWT) by offering a new perspective on the predictors of decent work. Originally, PWT posited work volition and career adaptability as key predictors contributing to decent work. Subsequent research, however, challenged this assumption by indicating the insignificant relationship between career adaptability as a predictor and decent work as an outcome, and thus career adaptability is no longer considered in recent studies of PWT (Douglass et al., 2017; Duffy et al., 2020). According to the findings of this study, career crafting emerges as a viable alternative to career adaptability. While work volition acts as a psychological resource that builds confidence in career choices, career crafting offers an action-oriented approach that, combined with work volition, contributes significantly to achieving decent work. Thus, career crafting is identified as a promising variable that warrants further examination within the PWT framework. Therefore, the empirical discovery of career crafting as a predictor of decent work substantially enriches the PWT framework.

Fifth, this study adds knowledge on the advances of the conservation of resources (COR) Theory. This study’s findings demonstrated that career crafting and decent work are unique forms of resources for improving PERMA+4 and, thus, well-being, especially in the absence of

substantial economic resources. This perspective broadens the application of COR theory within organizational psychology, suggesting that beyond the traditional focus on preserving existing resources, the active pursuit and development of new resources—such as those gained through career crafting—play a crucial role in promoting well-being. Career crafting involves activities that facilitate the acquisition of tangible and intangible assets, like relational resources, and strengthens an individual's internal resources. Understanding the role of the ongoing enactment in gaining resources adds knowledge to the conservation of resource theory.

Finally, this study expands the theoretical understanding of economic marginalization by introducing the concept of economic capital. This approach transcends simple income assessment by incorporating both past and future aspects of financial stability, thereby enriching the theoretical framework with a multidimensional view of economic resources. While this concept was employed to identify economically marginalized workers in this study, it holds potential for future application in examining statistical relationships with other measures, thereby broadening its utility in assessing both personal and work outcomes.

Practical Implications

This study offers insights for career practitioners and employee trainers in organizations on how to integrate aspects of career crafting and work volition into their programs as part of the ongoing practice. This study found that career crafting and work volition significantly and uniquely contribute to obtaining decent work and well-being. Career programs often focus on technical resources such as resume writing, interview techniques, searching information, and the like. They are important and well recognized, but the present study’s findings are evidence that to act out and develop career confidence are keys to greater access to decent work. Thus, research-driven practice should also consider helping personalized resources by participating in a

career network of one's interest, building confidence, building meaning out of hardship, and believing in growth potential. In particular, actively establishing relational career resources, gaining perspective by viewing where they currently stand as part of the journey to their career aspirations, and engaging in tasks aimed at advancing skills are key checkpoints for their clients. For instance, career development facilitators could prompt participants to share standout experiences from networking events, discuss the insights gained from these interactions, and plan their next steps or follow-up actions. By incorporating career crafting and work volition into their practice, career practitioners can increase the effectiveness of the program in measuring their clients' access to decent work and well-being.

Additionally, it is crucial for employers to understand that diversity extends beyond traditional categories and includes economic marginalization. This study revealed that heightened economic marginalization can impede career crafting and work volition, suggesting that EM workers may have weaker professional networks, skepticism towards career advancement, and diminished confidence. Acknowledging these unequal starting points between EM and non-EM workers is essential for developing practical support strategies. Career crafting and work volition are adaptable and can be taught. Therefore, to address these disparities, career practitioners can teach individuals how to actively engage in mentorship opportunities, build career confidence, and cultivate career meaning. Additionally, encouraging them to take on extra tasks that may enhance their abilities is beneficial. This approach will enable them to access better job opportunities and ultimately increase their satisfaction with the program. Similarly, employers can take proactive steps to bridge the gap in resources and opportunities to create a more equitable and inclusive workplace environment for all employees.

Finally, this study's findings suggest that EM workers and non-EM workers do use different mechanisms when experiencing well-being in a career-crafting context. While only three components of PERM+4—positive emotions, meaning, and growth mindset—were critical for EM workers, all nine components were critical for non-EM workers. This means that resources are limited, and thus, the absence of these would not be replaceable by other factors in EM workers. Given that, it is very important to cultivate positive emotions, meaning, and growth mindset by helping them act out career crafting. Measuring these three could also be critical for the workers not to be stressed out when career crafting but to enjoy the process, cultivating positive states and thus well-being.

One of the criticisms of vocational psychology is that its focus on meaning and fulfillment is seen as more relevant to the privileged population (Blustein, 2006). However, this study's findings reversed this perspective. Indeed, meaning could be more critical for EM workers than non-EM workers, alongside positive emotions and a growth mindset. Non-EM workers have other mediums to find happiness such as positive relationships, positive accomplishments, positive engagement, economic security, positive health, and positive work environment, in addition to these three elements, meaning that they have a variety of compensable resources to experience well-being, whereas EM workers hold meaning more importantly as one of three resources to well-being. This finding is particularly insightful for practitioners because it changes perspectives. Helping clients find personal meaning in their career journey is equally important for both EM and non-EM workers, but probably more important for EM workers from the perspective that other paths to well-being are limited to them.

Strengths and Limitations

This study has several methodological strengths and limitations. In terms of design, this study utilized a cross-sectional correlational design. Correlational design is suitable for the study's objective of analyzing the strengths and patterns of relationships between variables (Crano et al., 2011; Spector, 2019). However, the findings cannot claim causal relationships between the tested variables.

In terms of participants, this study utilized a crowdsourcing platform. The participant pool of crowdsourcing is diverse in terms of occupations, job roles, and work experience, and it closely mirrors the proportionate balance in gender and ethnicity, thereby providing a dataset that resembles the general population for analysis (Behrend et al., 2011; Buhrmester et al., 2011). For income, although additional data from lower-income workers had to be collected to reach the required sample size for two groups, the proportion of low-income workers in the final sample also closely resembled that of generally classified low-income workers (Brown et al., 2020; Loprest et al., 2009).

It should be noted, however, that the use of convenience sampling methods, such as crowdsourcing, limits the generalizability of the study findings. Specifically, this study might not have fully reached marginalized workers, potentially excluding those who are unfamiliar with such advanced technology. Additionally, workers from minority groups, for whom English language proficiency poses a barrier, were likely underrepresented, even though they represent important segments of marginalized populations. Therefore, caution should be exercised when generalizing the results to the broader marginalized population.

In terms of procedure and measure, the survey method was carefully developed following multiple cross-examinations with peer researchers and was deployed after review by the Institutional Review Board (IRB). The order of the questionnaire items was randomized to

prevent priming effects. Further, this study utilized standardized measures validated in previous research. These measures exhibited adequate internal consistency in this study's data, with the exception of two subfactors, thereby initially ensuring the reliability of the findings. Furthermore, this study triangulated economic resources by taking perspectives from Donaldson et al. (2021) and Duffy et al. (2016). This approach provided more confidence in classifying economic marginalization more effectively than using only income.

In addition, this study utilized self-reported data. Self-reported data may introduce potential biases affecting effect sizes (Donaldson & Grant-Vallone, 2002). Although self-reported data are generally reliable (Diener et al., 2002; Donaldson et al., 2021), and the non-sensitive nature of this study's variables would minimize the impact of social desirability bias (Krumpal, 2013), interpretation of the study findings should be cautious.

In terms of data analysis, this study employed a robust analytical framework to rigorously test the relationships among the variables of interest, utilizing advanced statistical techniques to ensure the accuracy of the study findings. I conducted a confirmatory factor analysis to assess the measurement model (i.e., economic capital) and a multi-group path analysis to test the consistency of the relationships among the study variables across two distinct groups (i.e., economically marginalized workers and non-economically marginalized workers), which provides initial applicability of the findings to a broader context. To deepen our understanding of the mechanisms and interactions among variables, I employed mediation and moderation multiple regression analyses. These analyses provided the direct and indirect effects among the variables, offering a nuanced view of how and under what conditions these relationships hold. Together, these advanced statistical analyses offer a comprehensive and nuanced examination of the relationships among variables. Again, these results indicate correlations, not causation.

Finally, while this study exhibits strengths commonly associated with quantitative research, due to the absence of qualitative data, rich content was not available, which might limit the depth of interpretation.

Future Directions

Longitudinal and intervention studies should be the next steps, with a particular focus on developing and examining career crafting intervention studies in the future. The combination of these studies with work volition is anticipated to have a stronger effect on well-being (Duffy et al., 2016). Investigating the effect of this combined intervention on PERMA+4 at different time points, especially concerning the three elements of PERMA+4 identified in this study—meaning, positive emotions, and a positive mindset— will be critical for a deeper understanding of career crafting's impact on PERMA+4 and overall well-being. Suggested meaningful time lag is six months when workers' well-being changes among the work-related constructs (Makikangas et al., 2016). While the future impact on decent work from career crafting actions may take time to manifest, improvements in PERMA+4 could be realized relatively quickly, given that the process of career crafting may generate resources for PERMA+4. This approach will not only provide practical insight into how to run a career crafting program but is also crucial for advancing our understanding of these relatively new concepts.

It will be beneficial for future studies to examine the effect of career crafting in rapidly evolving job markets, particularly given the increasing trend of remote work. Notably, in the U.S., 17% of working adults are fully remote, with a significant statistic revealing that 42% of those earning under $50,000 per year fall into this category (Mehdi & Sierra, 2024). The markedly high percentage of remote work among economically marginalized (EM) workers could potentially alter the landscape of employment patterns and, consequently, influence their

career development strategies. If marginalized workers are at the forefront of adopting remote work, understanding the influencing factors—such as their need for flexible work arrangements or their limited access to higher-paying, on-site job opportunities— will help to elucidate the broader socioeconomic and organizational contexts affecting the adoption of remote work and its implications for career crafting within EM workers.

Future research should prioritize the development of inclusive strategies that minimize language and technological barriers, thereby ensuring a broader and more representative understanding of marginalized groups' needs and experiences. To reach working adults less exposure to technology, it is possible to design community-based participatory research that involves local organizations, community leaders, and face-to-face interactions. These methods can include paper surveys, interviews, and focus groups conducted in community centers or other accessible locations. Engaging with communities directly allows researchers to gather data from those who may not have regular access to digital platforms. To reach language minorities with language barriers, it is possible to design research tools and methodologies that incorporate multilingual resources and translators. Additionally, leveraging visual and audio materials that transcend linguistic limitations can also be effective. Such efforts are essential for enhancing the validity and applicability of this study's findings to broader marginalized workers.

Incorporating additional outcome variables directly related to performance is crucial for future research. Organizations might not be fully satisfied with a focus solely on well-being, despite substantial research indicating significant correlations between well-being and performance (e.g., Donaldson et al., 2022). Moreover, a recent study reported a positive relationship between career crafting and performance (Chifor & Oprea, 2023). Nonetheless, organizations may seek more comprehensive insights, particularly understanding whether

increased engagement in career crafting benefits organizational success. Moreover, organizations have a fear of losing their employees. Talent retention is an organization's key focus area due to huge costs associated with talent attrition, such as recruitment costs and negative impacts on organizational revenue (Deepika & Sampurna, 2018; Vigoda, 2000). Career crafting may produce biases in association with turnover, which haven't been studied yet. Career crafting is positively related to work engagement (Lee et al., 2021), and work engagement is one of the most effective tools for ensuring employee retention (Deepika & Sampurna, 2018). However, there have been no studies that directly examined its relationship to turnover intention. Therefore, examining turnover intentions alongside performance as outcome variables for career crafting could help organizations adopt a more positive stance towards facilitating career development initiatives, although vice versa is always possible. Whatever the findings, career growth is an innate human aspiration as Maslow (1943) asserted in his Hierarchy of Needs theory. Workers, particularly Millennials and Gen Z, do not hesitate to pursue better opportunities that promise greater happiness, and organizations should recognize this as a natural progression for humanity. As such, continuous efforts to find a balance between organizational needs and employee satisfaction are essential. Thus, this type of future research could provide valuable insights for organizations to design more effective programs for their employees, aiming for mutual benefits.

Focusing on PERMA+4 within the specific context of careers offers a promising avenue for future research, potentially broadening our understanding of PERMA+4 in EM workers. PERMA interventions have been primarily focused on mental health to date; broadening attention toward particular career fields could be beneficial. Narrowing the scope to specific occupations, such as the arts or engineering, would allow for an in-depth examination of how

career actions tailored to these fields influence PERMA+4. Similarly, investigating career crafting and its relationship to decent work is another potential avenue for future research. This approach could offer valuable insights into how targeted career crafting within distinct sectors contributes to finding meaningful and fulfilling employment opportunities for EM workers.

Finally, future research should consider mixed methods in collecting data. Qualitative data, such as interviews or open-ended survey responses, can illuminate the underlying reasons behind the observed patterns and relationships among the variables. Such data may also reveal the complexities of participant experiences, perspectives, and the socio-cultural factors influencing the study's outcomes. By integrating both quantitative and qualitative approaches, future research could offer a more holistic understanding of the phenomena under investigation.

Conclusion

Many decades ago, it was believed that higher happiness was correlated with economic prosperity. This perspective has greatly faded with the emergence of subjective well-being, which places greater emphasis on psychological states. Perhaps it is time to reconsider this outdated notion with fresh perspectives. This could involve adopting a more rigorous approach to measuring economic factors, such as utilizing initially validated economic capital in this study.

When searching for a pathway to address social justice by narrowing the well-being gap, decent work should be considered an important pathway. Society can offer career programs aimed at helping economically marginalized individuals engage in career crafting and develop work volition. Such a specific pathway can not only broaden opportunities for decent work but, more importantly, strengthen the psychological states of PERMA+4. This study provides an empirical foundation for advocating the integration of career crafting and work volition into career development initiatives targeted at economically marginalized workers. By doing so, the

moral obligation to provide decent work is upheld, while also contributing to the broader goal of promoting well-being. It is hoped that the insights garnered from this study will inspire a concerted effort amongst career practitioners to implement programs that effectively address the needs of economically marginalized workers, paving the way for a more inclusive and equitable society.

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Appendix A

Economic Constraints Scale (Duffy et al., 2019)

1. For as long as I can remember, I have had very limited economic or financial resources.
2. Throughout most of my life, I have struggled.
3. For as long as I can remember, I have had difficulties making ends meet.
4. I have considered myself poor or very close to poor most of my life.
5. For most of my life, I have not felt financially stable.

The Satisfaction with Life Scale (SWLS, Diener et al., 1985)

1. In most ways my life is close to my ideal.
2. The conditions of my life are excellent.
3. I am satisfied with my life.
4. So far I have gotten the important things I want in life.
5. If I could live my life over, I would change almost nothing.

The Decent Work Scale (Duffy et al., 2017)

Factor 1. Physically and Interpersonally Safe Working Conditions

1. I feel emotionally safe interacting with people at work.
2. At work, I feel safe from emotional or verbal abuse of any kind
3. I feel physically safe interacting with people at work.

Factor 2: Access to Health Care

4. I get good healthcare benefits from my job.
5. I have a good healthcare plan at work.
6. My employer provides acceptable options for healthcare.

Factor 3: Adequate Compensation

7. I am not properly paid for my work (R).

8. I do not feel I am paid enough based on my qualifications and experience(R).

9. I am rewarded adequately for my work.

Factor 4: Hours that Allow for Free Time and Rest

10. I do not have enough time for non-work activities (R).

11. I have no time to rest during the work week (R).

12. I have free time during the work week.

Factor 5: Organizational Values Complement Family and Social Values

13. The values of my organization match my family values.

14. My organization's values align with my family values.

15. The values of my organization match the values within my community.

Career Crafting Assessment (CCA, Lee et al., 2021)

1. I make an effort to get to know people whose career I admire.

2. I make connections with people who share my career interests.

3. I make connections with people who have the skills I want to develop in my career.

4. I build contacts with people in areas where I would like to work.

5. I seek professional coaching from those whose careers I admire.

6. I attend events that will help me explore different career paths.

7. I ask others to introduce me to people who can positively influence my career.

8. I ask others for feedback on my career development.

9. I think about the ways in which my career positively impacts my life.

10. I remind myself that my career has personal significance.

11. I reflect on the role my career has for my overall wellbeing.

12. I look at a career as a means of expressing myself.

13. I choose to take on additional tasks at work.

14. I take on extra tasks that contribute to my career even if I do not receive extra pay for them.

15. Added work responsibilities excite me when they are relevant to my career interests.

Positive Functioning at Work (PF-W, Donaldson & Donaldson, 2020)

Positive Emotions

1. I feel joy in a typical workday.

2. Overall, I feel enthusiastic about my work.

3. I love my job.

Engagement

1. I typically become absorbed while I am working on something that challenges my abilities

2. I lose track of time while doing something I enjoy at work

3. When I am working on something I enjoy, I forget everything else around me

Relationships

1. I can receive support from coworkers if I need it.

2. I feel appreciated by my coworkers.

3. I trust my colleagues.

4. My colleagues bring out my best self.

Meaning

1. My work is meaningful.

2. I understand what makes my job meaningful

3. The work I do serves a greater purpose

Accomplishment

1. I set goals that help me achieve my career aspirations

2. I typically accomplish what I set out to do in my job

3. I am generally satisfied with my performance at work

Physical Health

1. I typically feel physically healthy.
2. I am rarely sick.
3. I can typically overcome sources of physical distress (e.g., insomnia, injuries, and vision issues)
4. I feel in control of my physical health.

Mindset

1. I believe I can improve my job skills through hard work.
2. I believe my job will allow me to develop in the future.
3. I have a bright future at my current work organization.

Work Environment

1. My physical work environment (e.g., office space) allows me to focus on my work.
2. There is plenty of natural light in my workplace.
3. I can conveniently access nature in my work environment (e.g., parks, oceans, and mountains).

Economic Security

1. I am comfortable with my current income.
2. I could lose several months of pay due to serious illness, and still have my economic security.
3. In the event of a financial emergency, I have adequate savings.

Work Volition Scale (Duffy et al.,2012)

Factor one. Volition

1. I've been able to choose the jobs I wanted.
2. I can do the kind of work I want, despite external barriers.
3. I feel total control over my job choices.
4. I've learned how to find my own way through the world of work.
5. I feel able to change jobs if I want to.

Factor two: Financial Constraints

6. Due to my financial situation, I need to take any job I can find (*R*)
7. When looking for work, I'll take whatever I can get (*R*).
8. In order to provide for my family, I often have to take jobs I do not enjoy (*R*).
9. I don't like my job, but it would be impossible for me to find a new one (*R*)
10. The only thing that matters to me in choosing a job is making ends meet (*R*).

Factor three: Structural Constraints

11. I feel that outside forces have really limited my work and career options (*R*).
12. The current state of the economy prevents me from working in the job I want (*R*).
13. Negative factors outside my personal control had a large impact on my current career choice (*R*).
14. The jobs I would like to pursue don't exist in my area (*R*).

Appendix B

Economic Capital: Means, Standard Deviations, and Factor Loadings Among Items

Factors and Items	*M*	*SD*	Factor Loadings 1	2	3	*h2*
Factor 1. Present Income	3.69	1.70				
Annual income level	3.69	1.70				.30
Factor 2. Past Economic Constraints (items from Duffy et al., 2019)	3.77	1.82				
For as long as I can remember, I have had very limited economic or financial resources (R).	3.85	1.94	.91			.88
Throughout most of my life, I have struggled (R).	3.88	1.96	.97			.90
For as long as I can remember, I have had difficulties making ends meet (R).	3.65	1.93	.91			.86
I have considered myself poor or very close to poor most of my life (R).	4.53	1.96	.88			.77
For most of my life, I have not felt financially stable (R).	3.97	1.98	.89			.80
Factor 3. Future Economic Security (items from Donaldson & Donaldson, 2020)	3.86	1.84				
I am comfortable with my current income.	3.57	2.11		.89		.74
I could lose several months of pay due to serious illness, and still have my economic security.	4.07	2.08		.95		.91
In the event of a financial emergency, I have adequate savings.	3.94	1.85		.71		.62

Note. N=629. Factor 2 comprises items from the lifetime economic constraints scale (Duffy et al., 2019) and is labeled as past economic constraints in this study. Factor 3 comprises items from economic security (Donaldson & Donaldson, 2020) and is labeled as future economic security in this study. All were measured using 7-point scales. Income was re-coded from the originally collected 16 levels into seven levels. Each item was loaded solidly to one of three factors, ranging from .71 to 1.00 and in terms of commonality, the income item showed relatively low communality (.30) but acceptable when .35 is recommended for retaining items (Costello & Osborne, 2005).

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